MIS 5394 : MSBA Capstone
Final Presentation

Sharadha Varadharaj
Asitha Mudiyanselage
Grayson Arcemont

Baylor University

Project Overview

Objective

The goal is to perform a comprehensive analysis of the MCL dataset to obtain insights that might potentially enhance business decisions relating to healthcare services.

Data Description

Our dataset initially comprised 48 variables and ***115,371 observations***. For our analysis, we selected ***17 key variables***. Before data cleaning, we identified the following number of missing values in each of these variables.

```
> variables_names
 [1] "Member ID"                   "Plan Effective Date"         "Group ID"
 [4] "Date Received"               "Date of Service"             "Provider Type...6"
 [7] "Claims Bucket"               "BC"                          "Patient Responsibility"
[10] "DA"                          "RDHRes"                      "MC"
[13] "Deduct"                      "Pricing Strategy"            "Pre-Authorization"
[16] "Assigned"                    "Status"                      "Denial Reason"
[19] "Completed date MM/DD/YYYY"   "Co-Ins"                      "QPA Amount"
[22] "CHECK #"                     "Check Date"                  "Date(s) of Service:"
[25] "Claim ID...25"               "Provider First Name:"        "Provider Last Name"
[28] "Provider Type...28"          "Billing Provider Name:"      "Billing Provider Address:"
[31] "City"                        "State"                       "Zip Code"
[34] "Country"                     "CPT Codes:"                  "NPI Number:"
[37] "Place of Service"            "Tax ID"                      "Provider Type...39"
[40] "Claim ID...40"               "Luminx Provider Type Code"   "Luminx Specialty Code"
[43] "DCN"                         "visits"                      "DataIsight"
[46] "Coverage Tier"               "Medicare"                    "PHCS"
```

```
> sapply(MCL_data_selected, function(x) sum(is.na(x)))
                  Provider Type                                  Claims Bucket          BIlled Charges     Patient Responsibility
                              5                                            285                      90                     112549
Discount Amount (Saving amount)                             RDH Responsibility            Member Copay                 Deductible
                            475                                          36663                   83138                      98817
               Pricing Strategy                              Pre-Authorization                  Status                 QPA Amount
                          23740                                             62                       4                     115216
                       # visits 120% of medicare or the equivalent DataIsight           Coverage Tier              100% Medicare
                            211                                          22999                     599                      47557
                           PHCS
                          58917
```

```
 [1] "Provider Type"                     "Claims Bucket"
 [3] "BIlled Charges"                    "Patient Responsibility"
 [5] "Discount Amount (Saving amount)"   "RDH Responsibility"
 [7] "Member Copay"                      "Deductible"
 [9] "Pricing Strategy"                  "Pre-Authorization"
[11] "Status"                            "QPA Amount"
[13] "# visits"                          "120% of medicare or the equivalent DataIsight"
[15] "Coverage Tier"                     "100% Medicare"
[17] "PHCS"
```

```
> qualitative_vars
[1] "Provider Type"     "Claims Bucket"     "Pricing Strategy" "Pre-Authorization" "Status"       "Coverage Tier"     "PHCS"
> quantitative_vars
 [1] "BIlled Charges"                                 "Patient Responsibility"
 [3] "Discount Amount (Saving amount)"                "RDH Responsibility"
 [5] "Member Copay"                                   "Deductible"
 [7] "QPA Amount"                                     "# visits"
 [9] "120% of medicare or the equivalent DataIsight"  "100% Medicare"
```

Baylor University

Data Cleaning

Handling Missing Values

Provider Type

- Issue: Low-frequency categories and inappropriate data types.
- Handling: Consolidated **low-frequency** categories, converted variables to appropriate data types.

Claims Bucket

- Issue: Multiple categories with significant missing values and inconsistencies.
- Handling: Addressed missing values, **consolidated categories**, converted to a factor.

Patient Responsibility

- Issue: Significant deviation from normality with clustered values.
- Handling: Imputed missing values with the **median** due to skewed distribution.

Discount Amount

- Issue: Right-skewed distribution with high variance.
- Handling: **Median** imputation for missing values, handled high variance.

RDH Responsibility

- Issue: Right-skewed distribution with several outliers.
- Handling: Imputed missing values with the **median**, addressed skewness

QPA Amount

- Issue: Highly skewed distribution with most values at 1.00.
- Handling: Imputed missing values with **0**.

Billed Charges

- Issue: Significant deviation from normality, right-skewed distribution with outliers.
- Handling: **Median** imputation for missing values, identified and handled extreme outliers to avoid skewing further analysis.

Member Copay

- Issue: Skewed distribution with outliers.
- Handling: Imputed missing values with **zero**, ensuring data completeness and consistency.

Of Visits

- Issue: Highly skewed distribution with most values at one visit.
- Handling: **Median** imputation for missing values, categorized data into binary and detailed groups for better analysis.

120% Medicare & 100% Medicare

- Issue: Right-skewed distribution with extreme outliers.
- Handling: **Zero** imputation for missing values, ensuring a conservative and consistent approach.

Baylor University

PHCS

- Issue: Wide range of values with significant variability.
- Handling: **Zero** imputation for missing values, assuming missing entries indicated a lack of PHCS data.

Deductibles

- Issue: Skewed distribution with significant missing values.
- Handling: **Imputed missing** values with **zero**.

Pricing Strategy

- Issue: Mixture of standard categories and numeric values.
- Handling: **Standardized categories**, imputed missing values with the **mode**.

Status

- Issue: Multiple similar categories and different capitalizations.
- Handling: **Consolidated** similar categories, imputed missing values.

Pre-Authorization

- Issue: Multiple variations of "Yes", "No", and "Unknown".
- Handling: **Standardized categories**, imputed missing values with the **mode**.

Coverage Tier

- Issue: Mixture of standard category labels, numeric values, and invalid entries.
- Handling: **Standardized categories**, imputed missing values with the **mode**.

Purpose of Different Types of Visualizations

Density Plots

It examines how to visualize the distribution of the Billed Charges and Discount Amount to help understand how these ***variables are spread*** across the dataset.

Bar Charts

It examines the ***distribution*** of the categorical variables and the types that lie within the data and go into the types that are significant within this cleaned dataset.

Box Plots

It reveals ***significant variations*** in RDH Responsibility across different categories, with log transformations helping to normalize the data and provide clearer comparisons. Outliers are a consistent feature, particularly in categories like Hospitals, Mutual claims, EMP and FAM coverage tiers, Complete status, and Medicare pricing strategy.

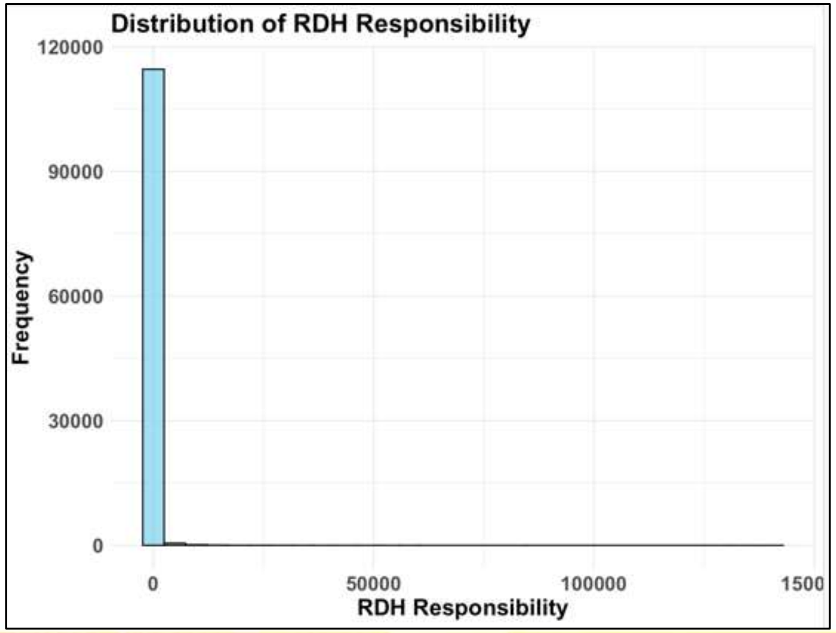
Correlation Matrix

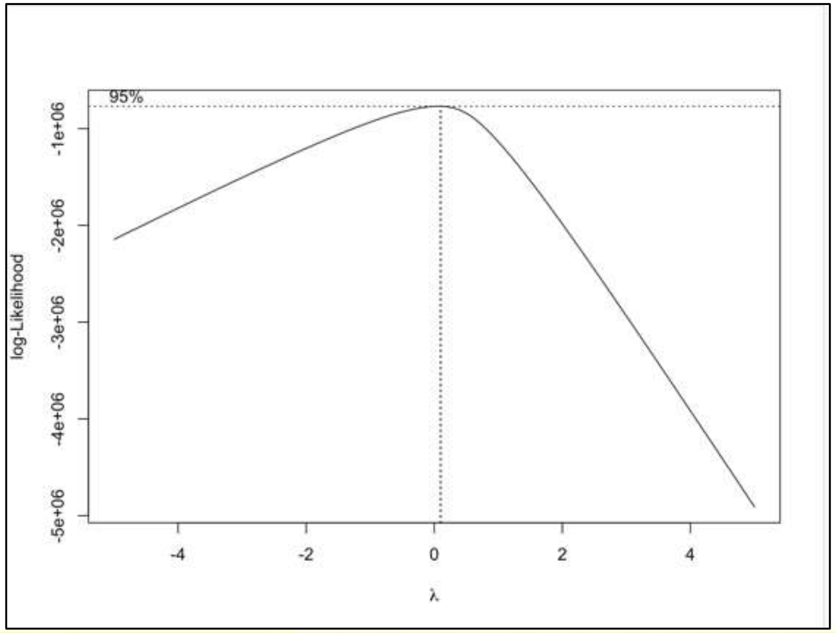
It shows the strongest correlations exist between billed charges, discount amounts, and Medicare-related values, indicating these ***variables are closely linked*** and should be considered together in further analysis

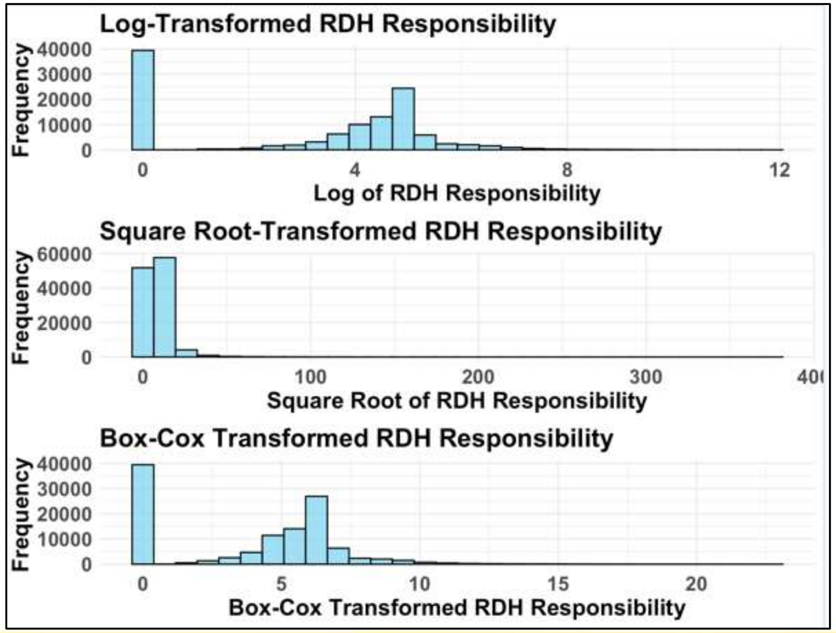
Visualization

RDH Responsibility:

- The original histogram shows a highly skewed distribution with data concentrated at lower values.
- A long tail extends towards higher values, indicating significant skewness.
- Log transformation slightly **normalizes** the distribution but still shows a right-skewed pattern.
- Square root transformation reduces skewness but retains some asymmetry, Box-Cox transformation, optimized using maximum likelihood estimation, offers the most normalized distribution.
- Overall, **log transformation** is the most effective in normalizing the distribution, though all transformations improve upon the original skewness to some extent.



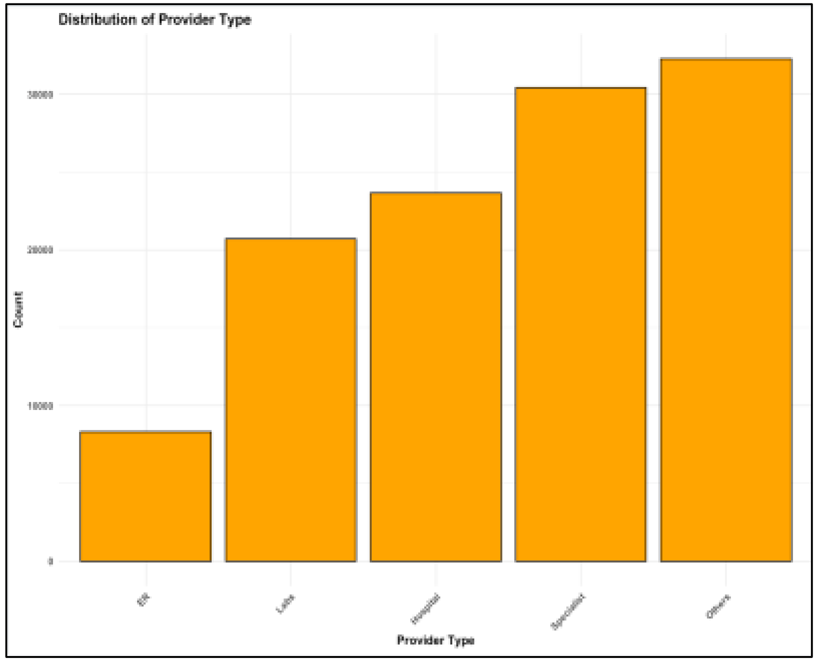


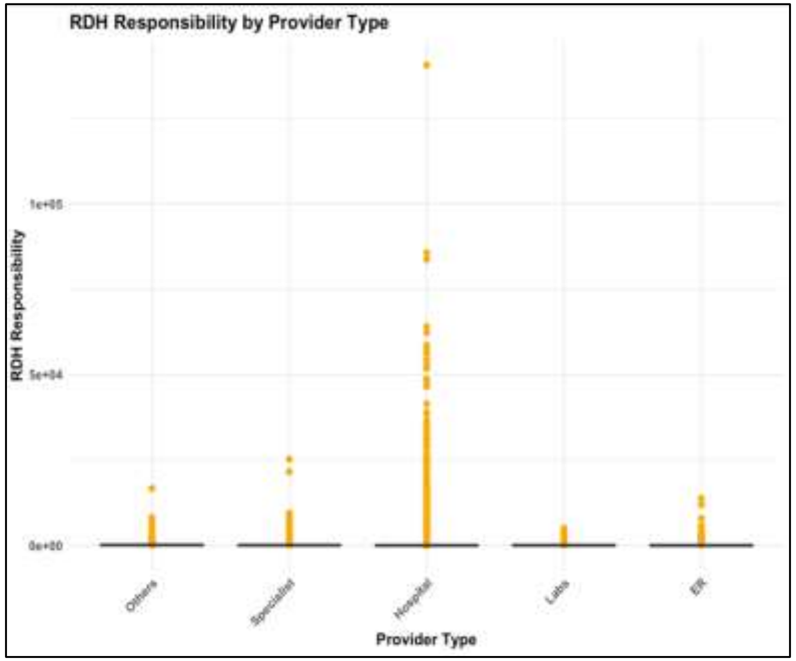


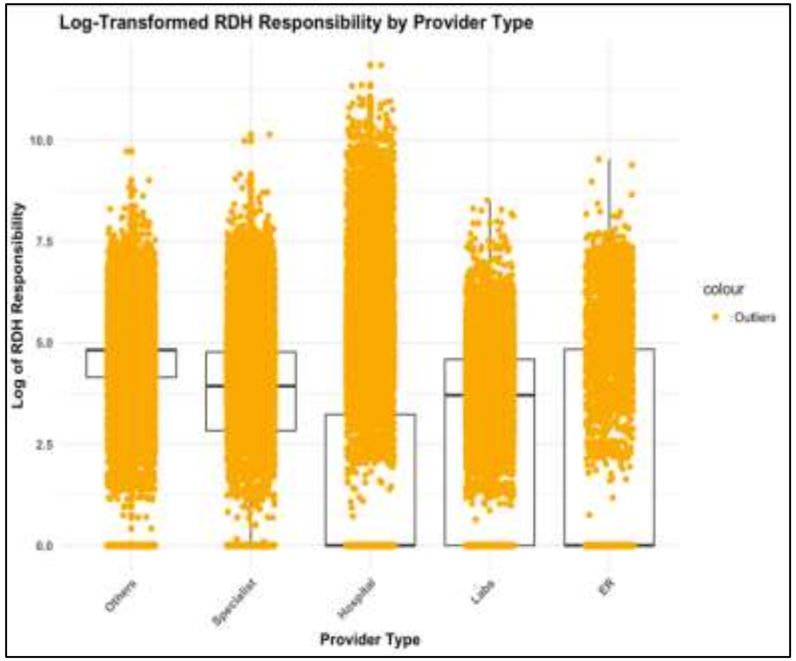
Baylor University

Provider Type:

- The distribution of RDH Responsibility across different Provider Type categories shows **significant variation**.
- **Hospitals** have the widest spread and highest values of RDH Responsibility.
- Numerous **outliers** are present in the Hospital category.
- Outliers in the Hospital category are more evenly distributed on a log scale.
- Hospitals continue to show **higher responsibility ranges**, while Specialists, ER, Labs, and Others have moderate to lower ranges.

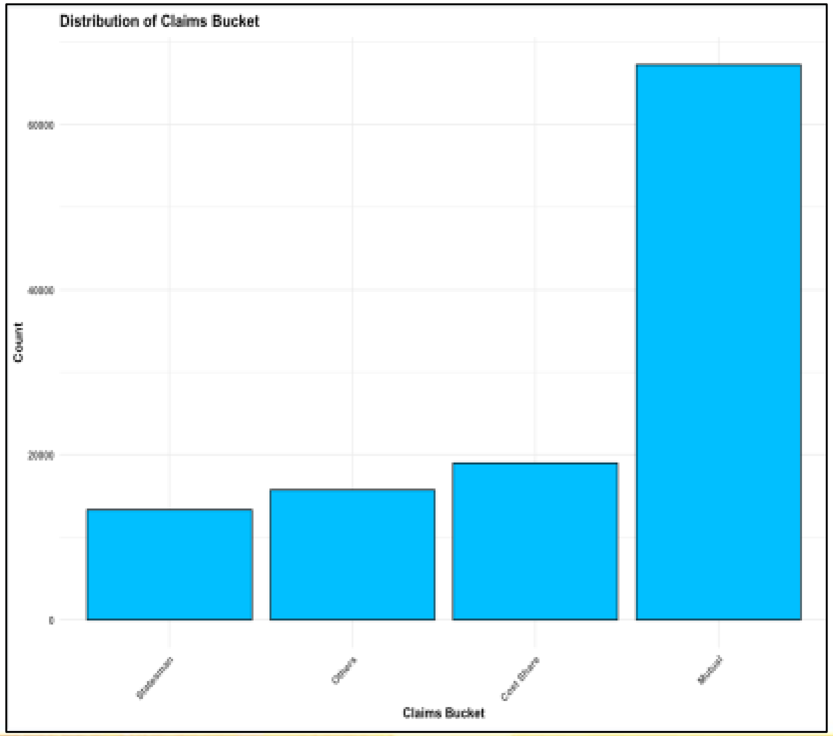


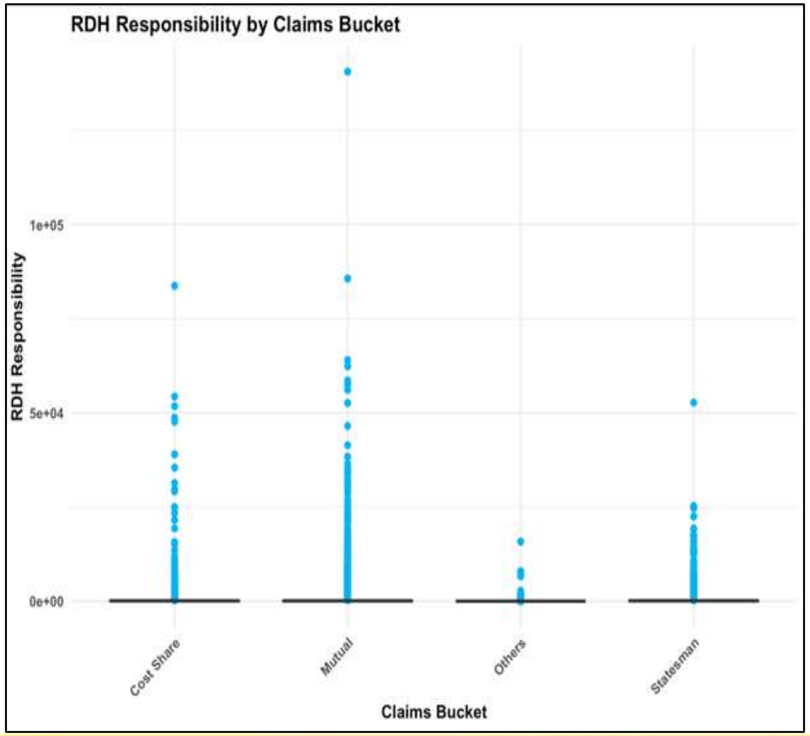


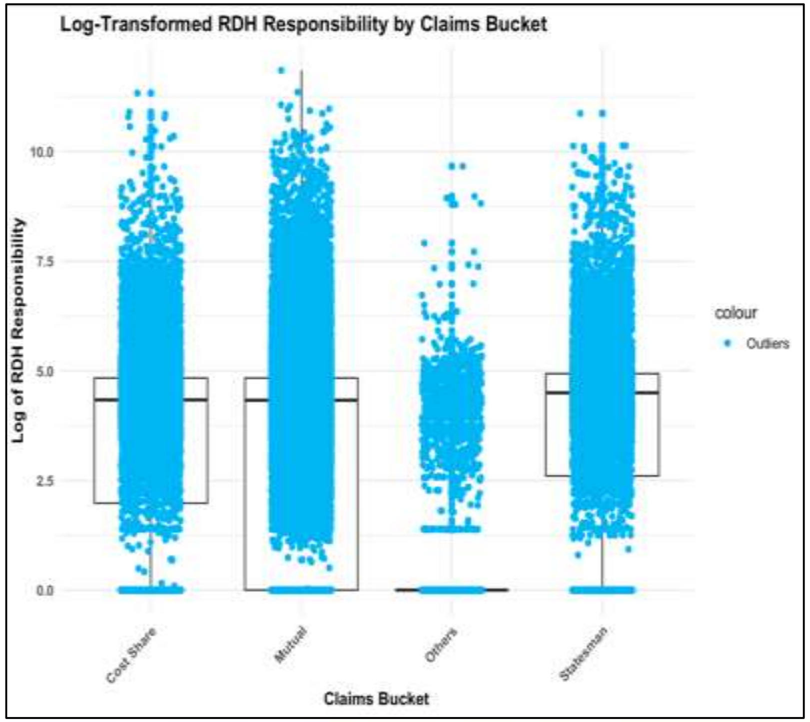


Claims Bucket:

- **Mutual** exhibits the widest spread of RDH Responsibility values with numerous high-value outliers.
- Cost Share and Statesman show a moderate spread with several high-value outliers.
- Others have a lower range of RDH Responsibility values with fewer and less extreme outliers.
- Outliers in the Mutual category are more evenly distributed on a log scale.
- Mutual continues to show **higher responsibility ranges**, while Cost Share, Statesman, and Others have moderate to lower ranges.

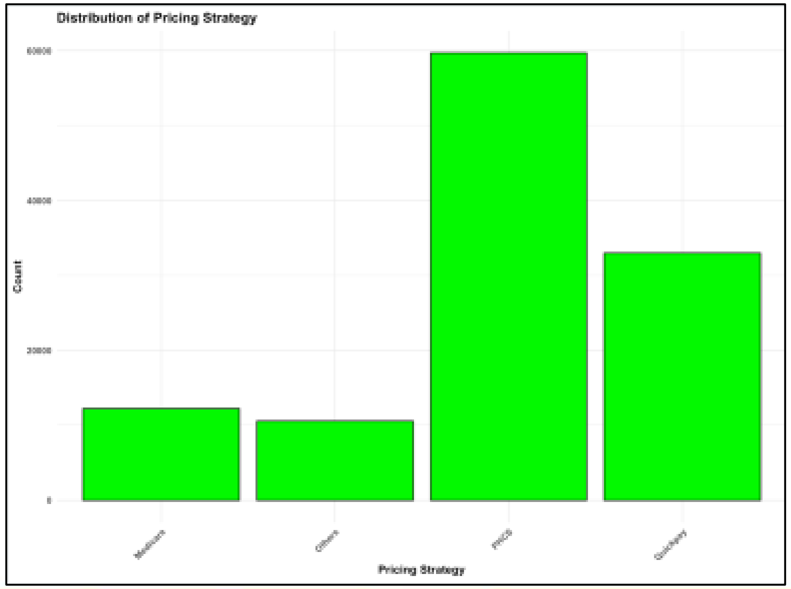


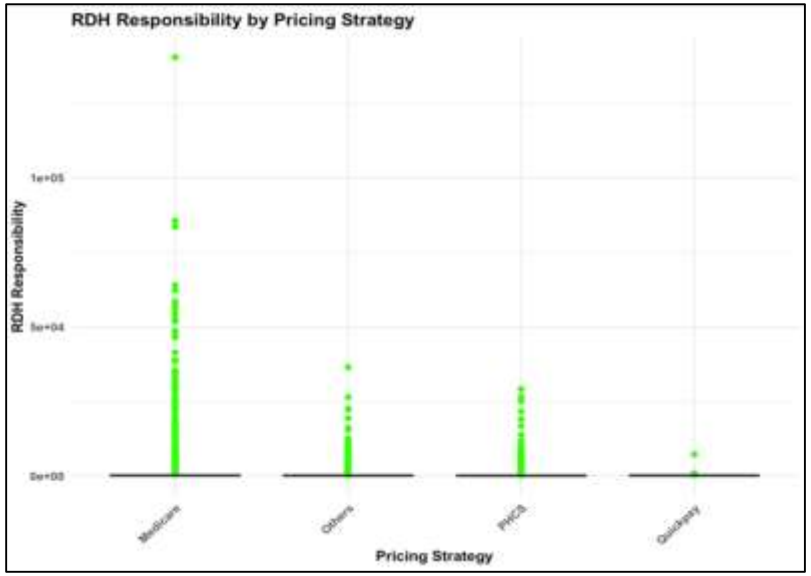


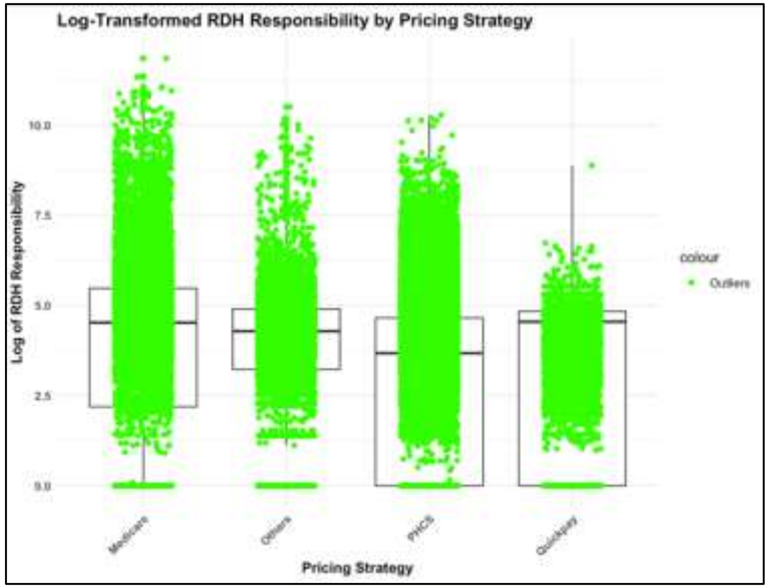


Pricing Strategy:

- The distribution of RDH Responsibility varies significantly across different Pricing Strategy categories.
- **Medicare** shows the highest values and widest spread of RDH Responsibility, indicating larger amounts owed.
- Others display a moderate range of RDH Responsibility with some high-value outliers.
- PHCS also shows a moderate range of responsibilities with occasional high-value outliers.
- Medicare still shows the **highest concentration** of outliers, but they are more evenly distributed on a log scale.
- Medicare **continues** to show a higher responsibility range, indicating significant variation in amounts owed even on a log scale.
- Others and PHCS show a moderate range of responsibilities with occasional high outliers.
- QuickPay has lower ranges of responsibilities with fewer and less extreme outliers.



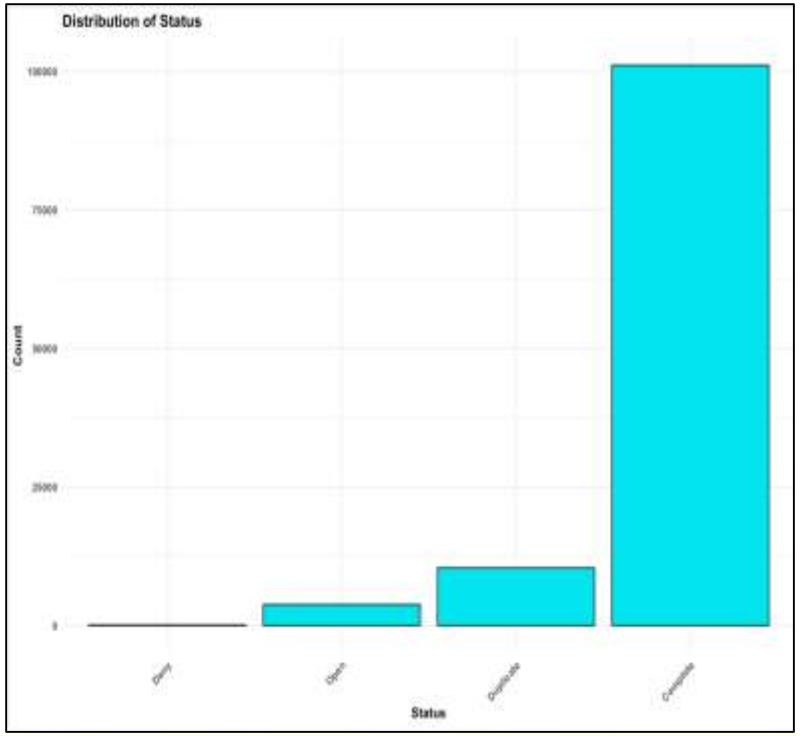


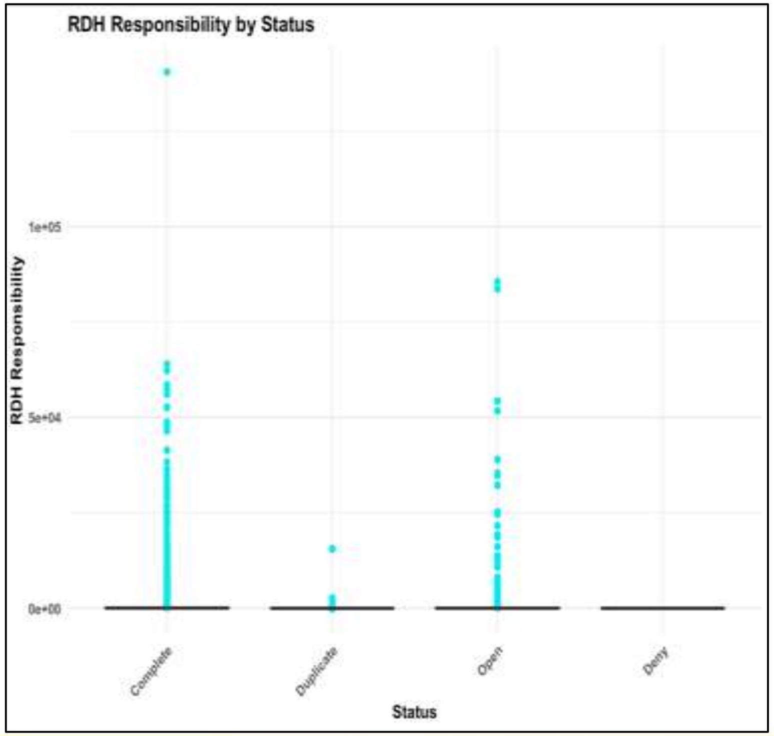


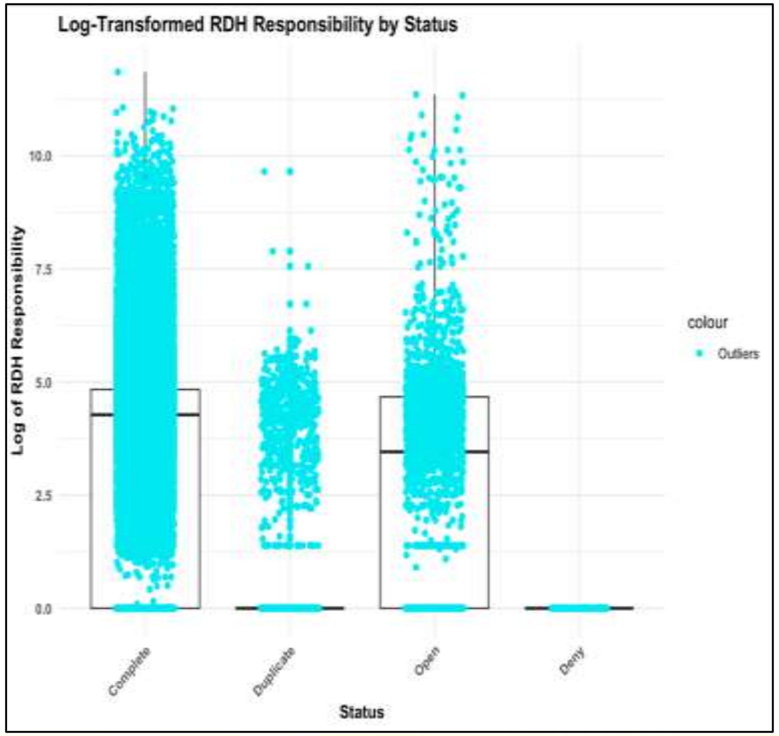
Baylor University

Status:

- The bar chart shows "Complete" as the most common status with nearly 100,000 entries, followed by "Duplicate" with around 10,000 entries.
- "Open" status are fewer, around 3,000, while "Deny" statuses are almost negligible.
- RDH Responsibility varies significantly across Status categories, with "Complete" showing the widest spread and several high-value outliers.
- "Open" statuses show a moderate spread with some high-value outliers, while "Duplicate" and "Deny" have lower ranges with fewer outliers.
- "Complete" still shows the highest concentration of outliers on a log scale, indicating significant variation in amounts owed.

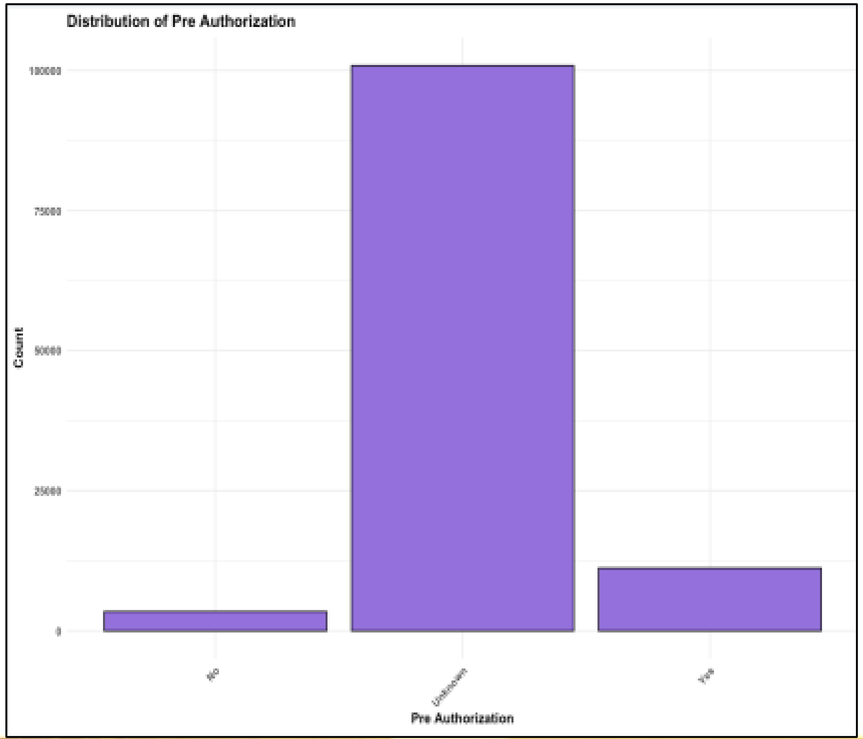


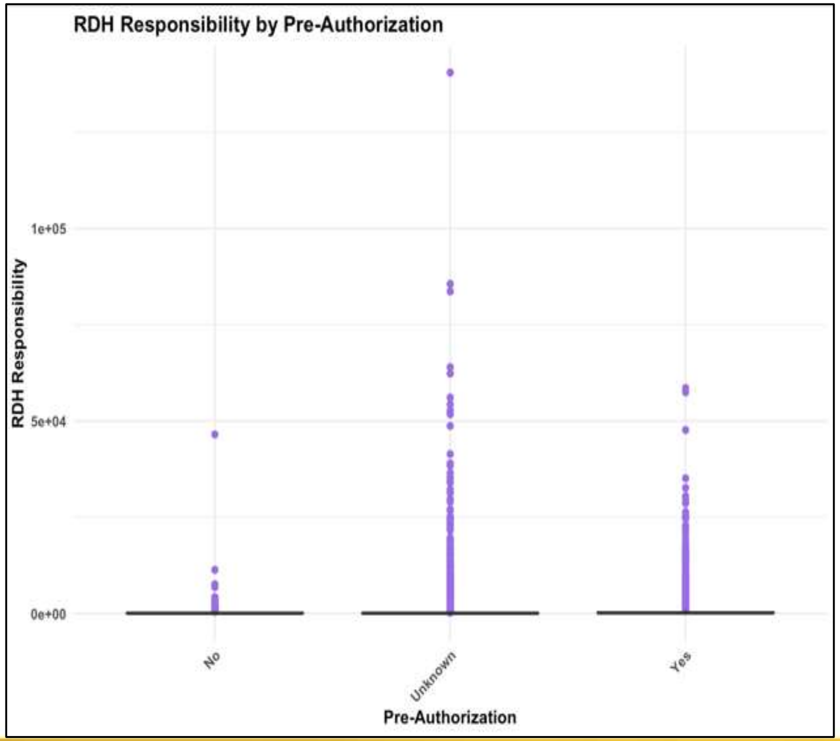


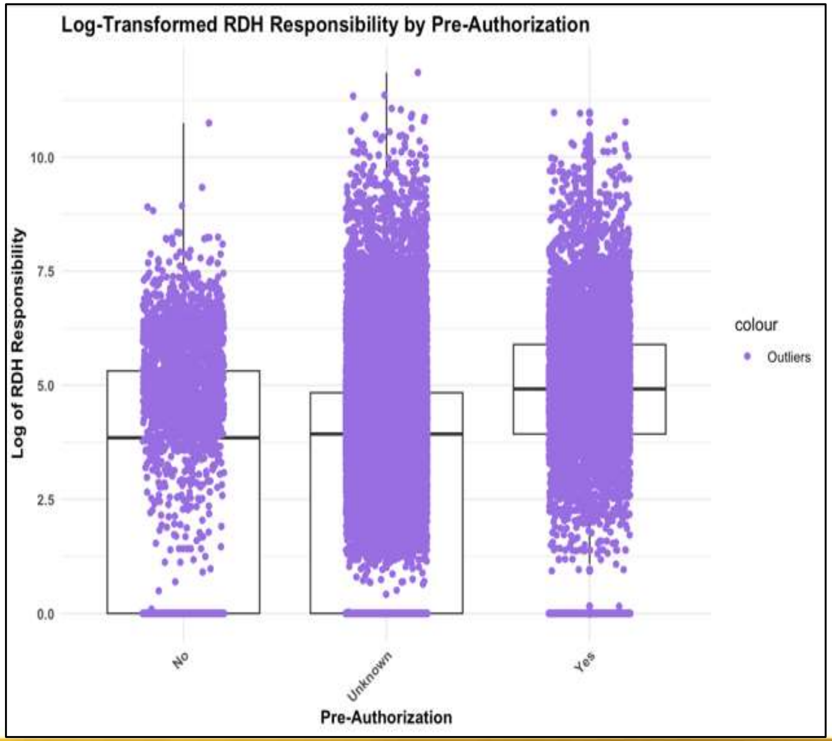


Pre-Authorization:

- RDH Responsibility varies significantly across Pre-Authorization categories, with "Unknown" showing the widest spread and several high-value outliers.
- "Yes" categories show a **moderate** spread with some high-value outliers, but not as extensive as "Unknown."
- "No" categories have a **lower** range of RDH Responsibility values with fewer and less extreme outliers.
- Jitter points for outliers show that "Unknown" still has the highest concentration of outliers, but they are more evenly distributed on a log scale.
- "Unknown" continues to show a higher responsibility range, while "Yes" has a moderate range and "No" has lower ranges with fewer outliers.

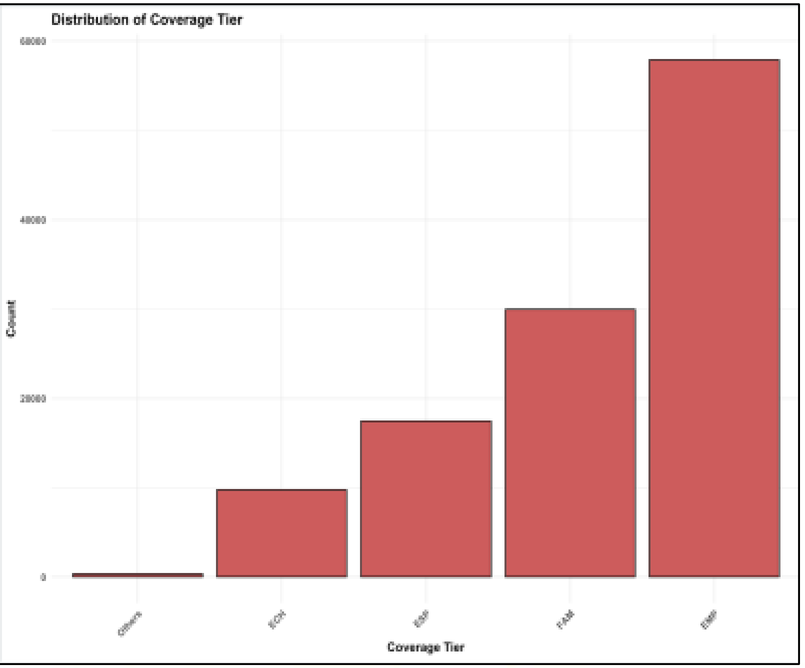


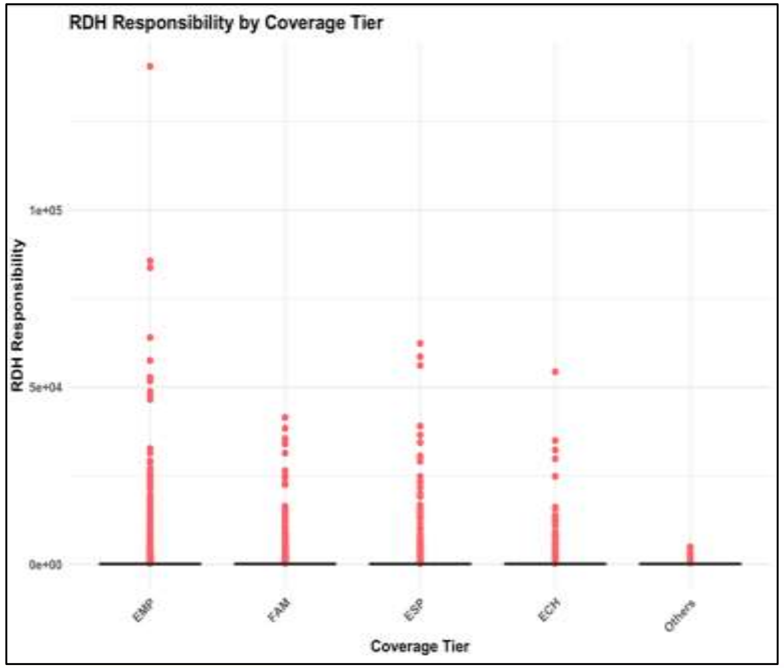


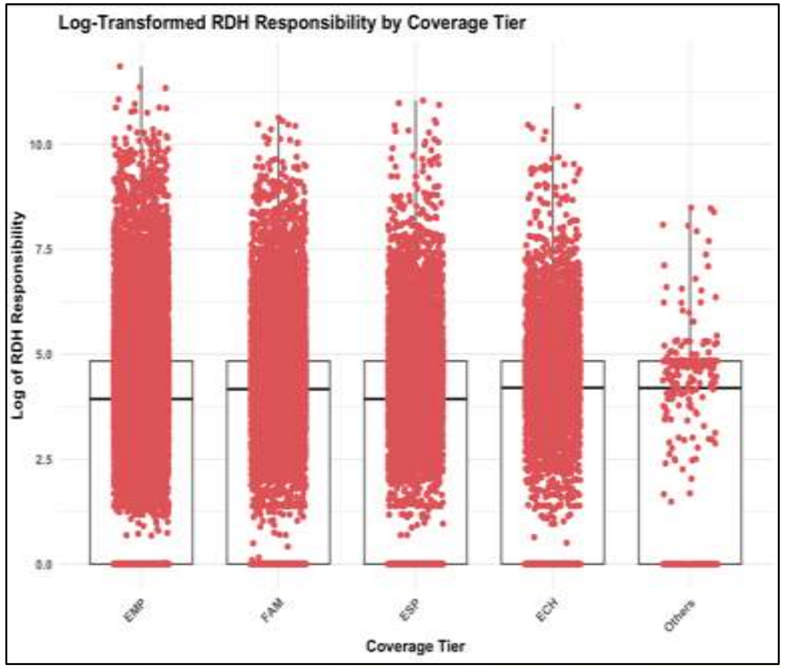


Coverage Tier:

- "EMP" (Employee) is the most common coverage tier with nearly 60,000 occurrences, followed by "FAM" (Family) with around 30,000 entries.
- "ESP" has about 20,000 entries, and "ECH" is the least common among the main categories with around 10,000 occurrences.
- The distribution of RDH Responsibility varies significantly across different Coverage Tier categories, with **EMP and FAM** showing the widest spread and highest values.
- Numerous outliers are present in the EMP and FAM categories, indicating many instances with exceptionally high RDH Responsibilities.
- EMP and FAM continue to show the hig**hest responsibility ranges** and concentrations of outliers, even on a log scale.
- ESP and ECH show a **moderate range of responsibilities**, with some outliers indicating occasional high responsibilities.
- Others has a **lower range of responsibilities** with fewer and less extreme outliers.

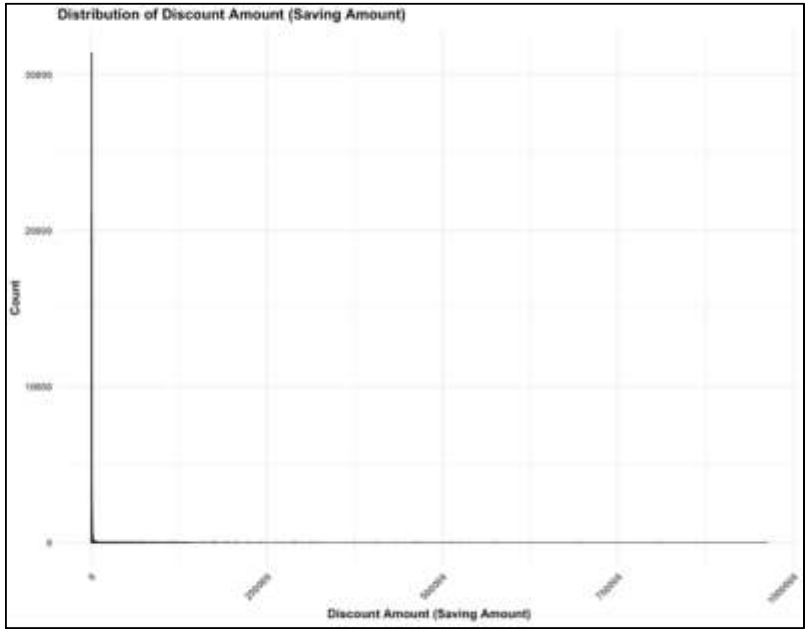


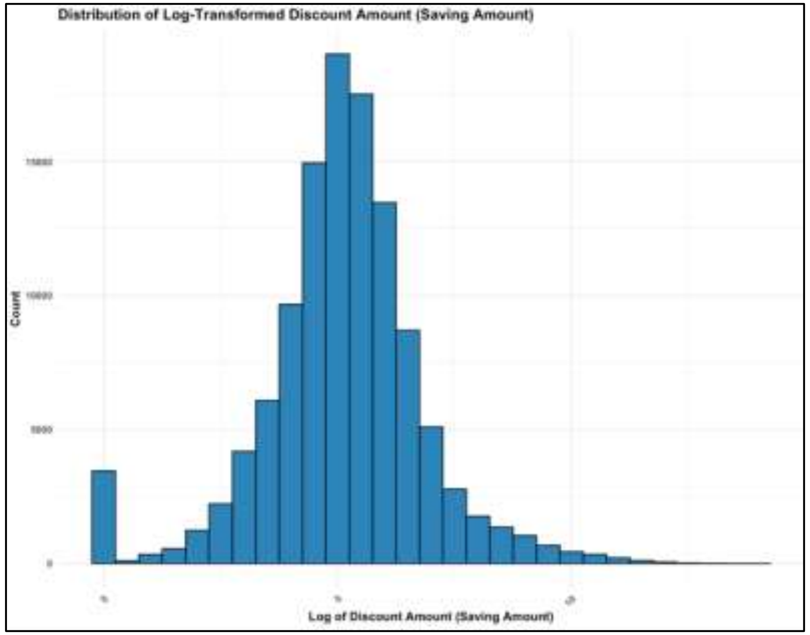




Discount Amount:

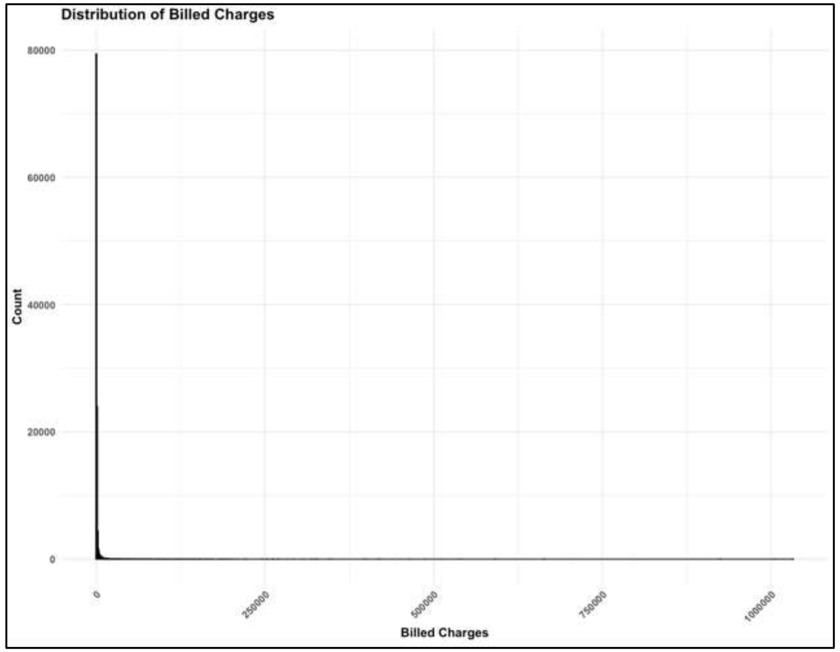
- The initial histogram shows a significant right skew with most values clustered near zero.
- A long tail extends towards higher amounts, indicating a highly uneven distribution.
- The skewness makes the initial distribution difficult to analyze and interpret.
- Log transformation normalizes the distribution, resulting in a more symmetrical, bell-shaped curve.
- The log-transformed data reduces skewness and highlights the central tendency.
- Log transformation handles wide-ranging data, making patterns more apparent and stabilizing variance for accurate statistical analysis and model building.

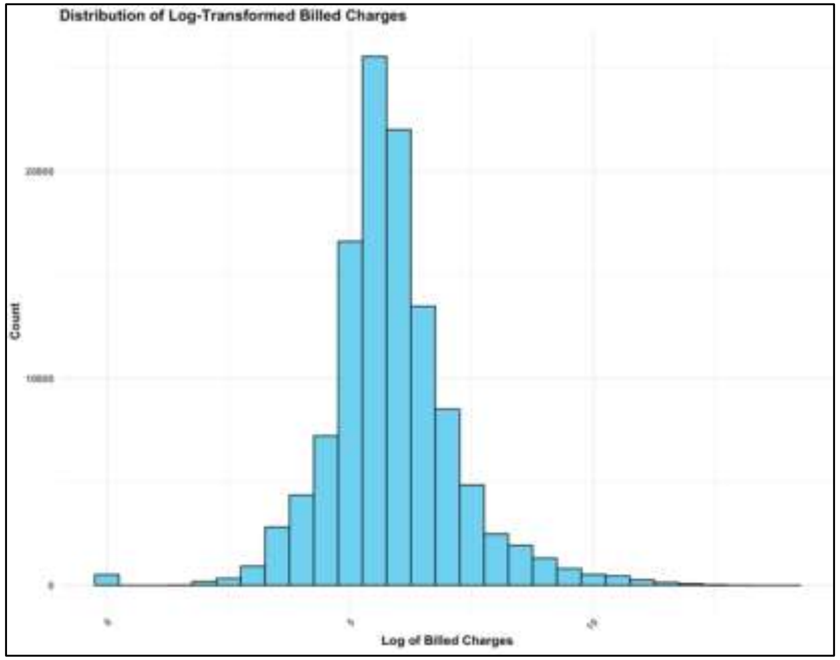




Billed Charges:

- The original histogram of Billed Charges shows a highly **right-skewed** distribution.
- Most values are clustered near the lower end, with a long tail extending towards higher values.
- Log transformation results in a much more **symmetrical** and normalized distribution.
- The transformation reduces skewness, bringing the data closer to a normal distribution.
- Log transformation compresses the range of high values, balancing the distribution and mitigating outliers.
- Improved interpretability, reduced skewness, robust statistical models, and better model performance.

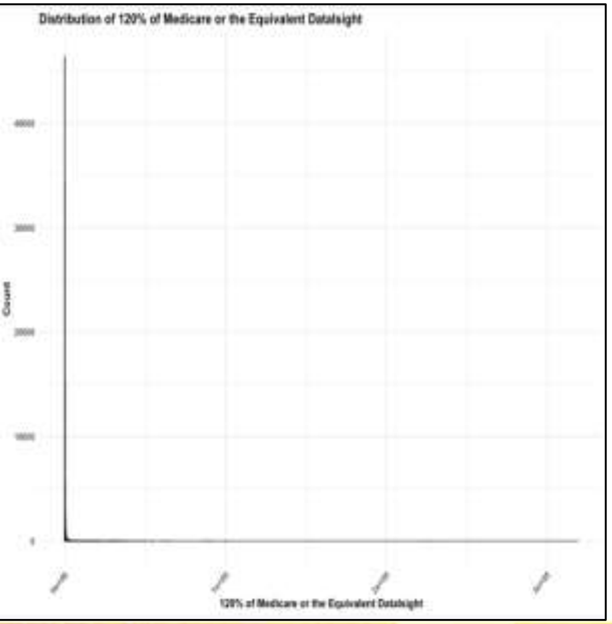


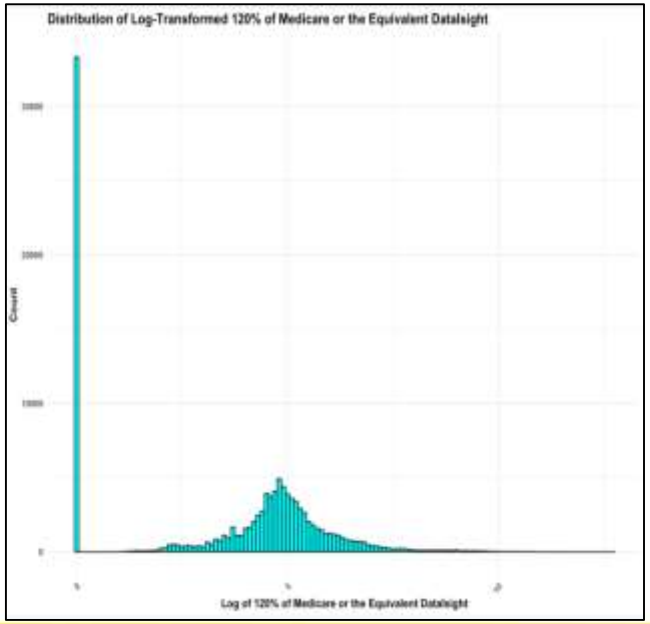


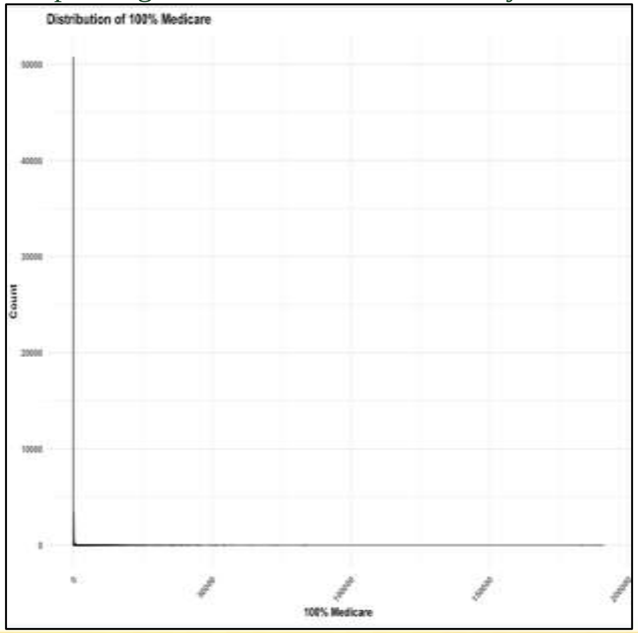
120% Medicare :

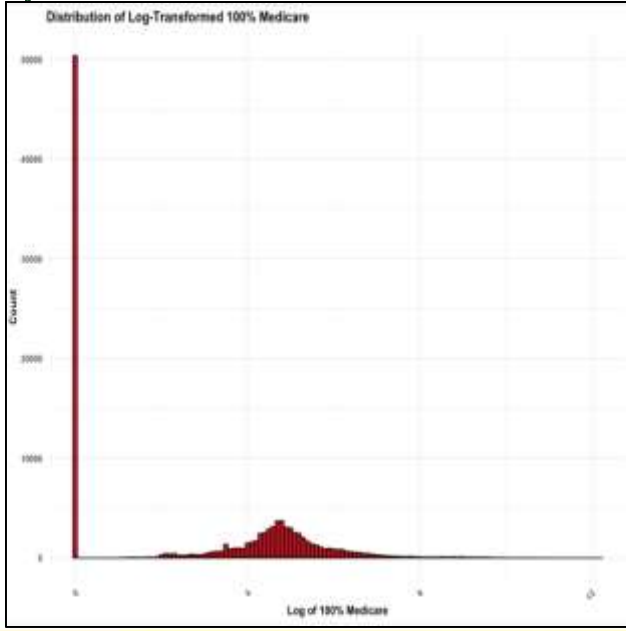
- The initial distribution of the variable shows a highly **skewed pattern** with most data concentrated at lower values.
- A long tail extends to higher values, indicating the presence of extreme outliers.
- The skewness makes it challenging to perform meaningful statistical analyses.
- The **log-transformed** data shows a more **normalized distribution**, reducing the impact of extreme outliers.
- The transformation stabilizes variance and makes the data more suitable for statistical modeling and analysis.

- *100% Medicare:*
- The original histogram of the 100% Medicare variable shows a highly **skewed distribution**, with most data concentrated near lower values.
- A long tail extends towards higher values, indicating non-normal distribution and potential issues for statistical analyses assuming normality.
- The **log-transformed** histogram reveals a more **normalized distribution**, with data spread more evenly across the range of values.
- The transformed data becomes more suitable for parametric tests and regression models, improving the robustness and accuracy of the analysis.



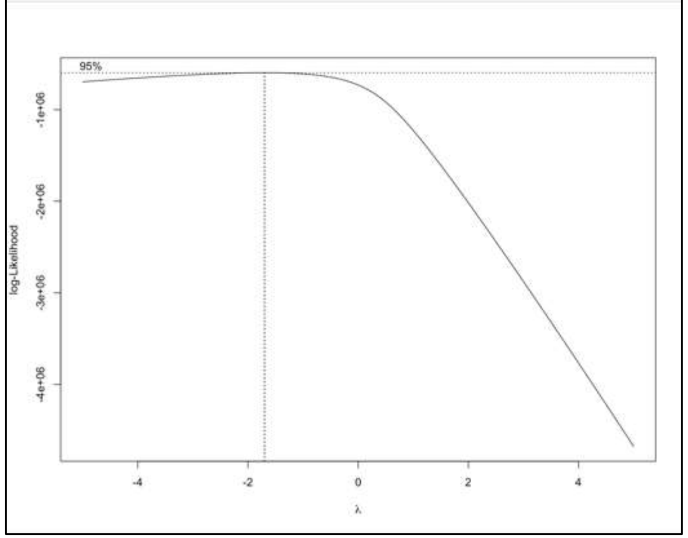


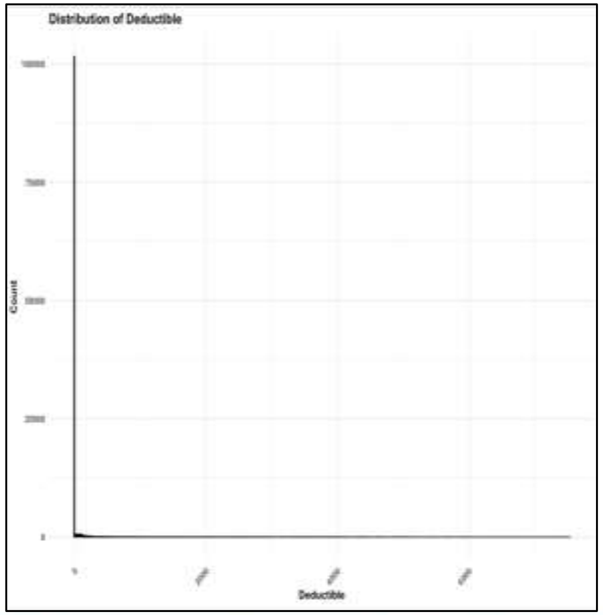


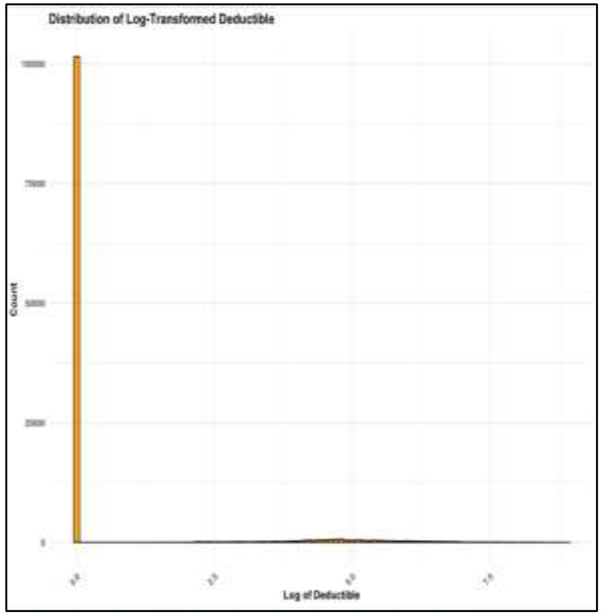


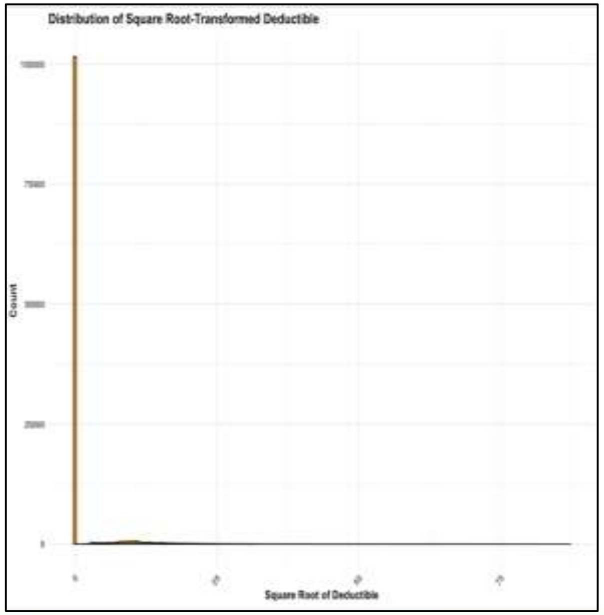
Deductibles:

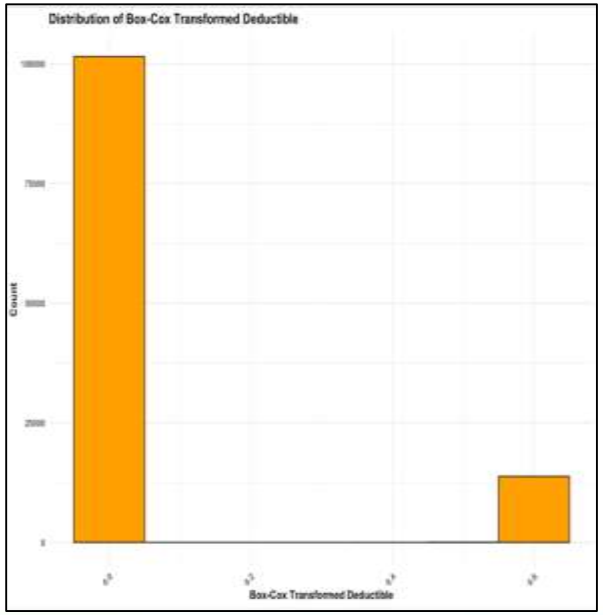
- The original distribution of the deductible variable is highly skewed with a significant number of zero values.
- Most counts are concentrated at the lower end of the deductible range.
- **Log transformation** was applied to handle zero values.
- The resulting distribution shows non-zero values spread out more evenly, but the overall distribution remains skewed.
- The **square root** transformation reduces skewness but still leaves a high concentration of zero values, the distribution remains far from normal.
- **Box-Cox** transformation was applied by adding a constant to make all values positive and finding the optimal lambda.
- Despite the transformation, the distribution still shows a significant concentration at zero, indicating **limited effectiveness**.





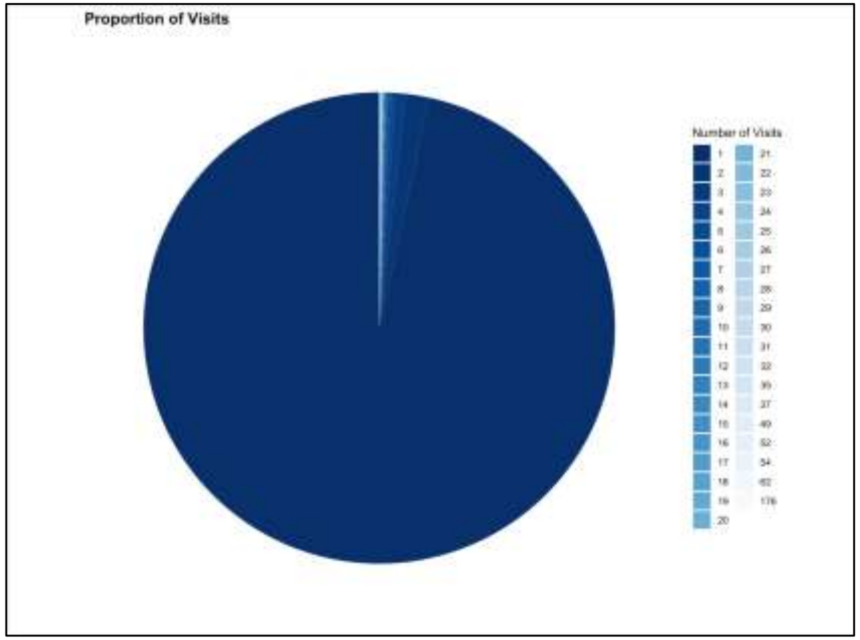


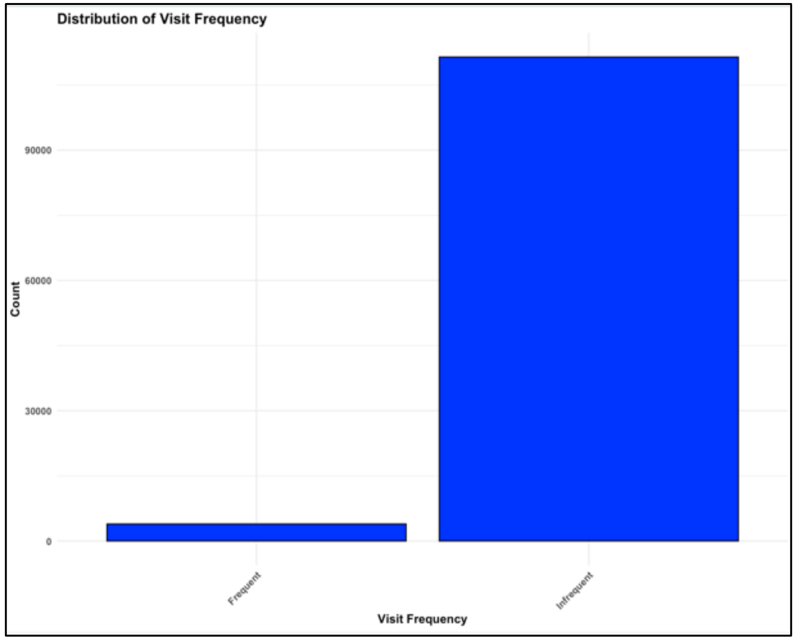


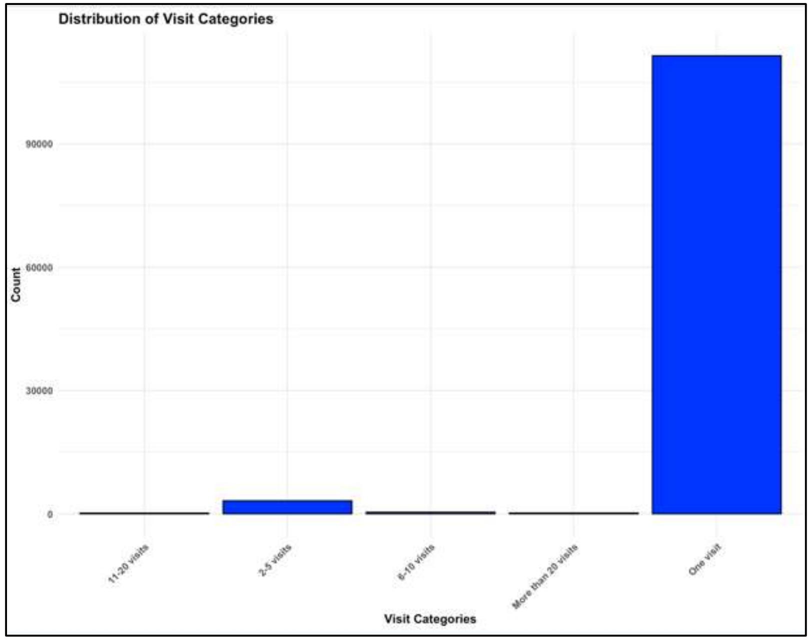


Visits:

- **Binary Transformation (visit binary):** Divides the data into "Frequent" (more than one visit) and "Infrequent" (one visit), showing a clear distinction between categories.
- **Distribution of Visit Frequency:** The majority of visits are categorized as "Infrequent", indicating most members have only one visit.
- **Categorical Grouping (visit category):** Groups data into "One visit", "2-5 visits", "6-10 visits", "11-20 visits", and "More than 20 visits".
- Most data points fall into the "One visit" category, with significantly fewer entries in higher visit categories.
- Transformations make the data more manageable and interpretable, aiding in the analysis of visit frequency trends and patterns.







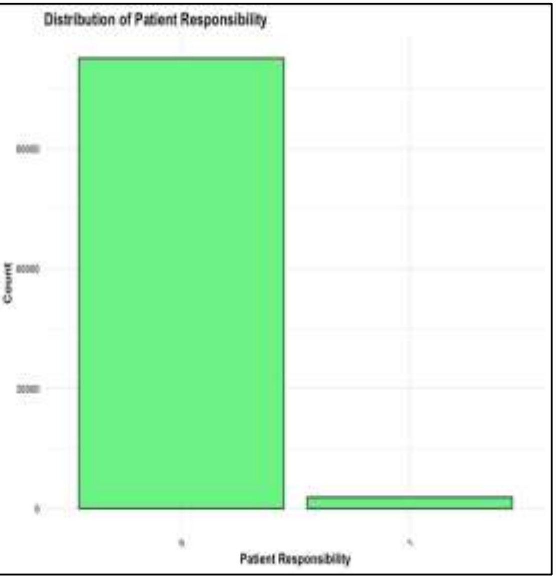
Baylor University

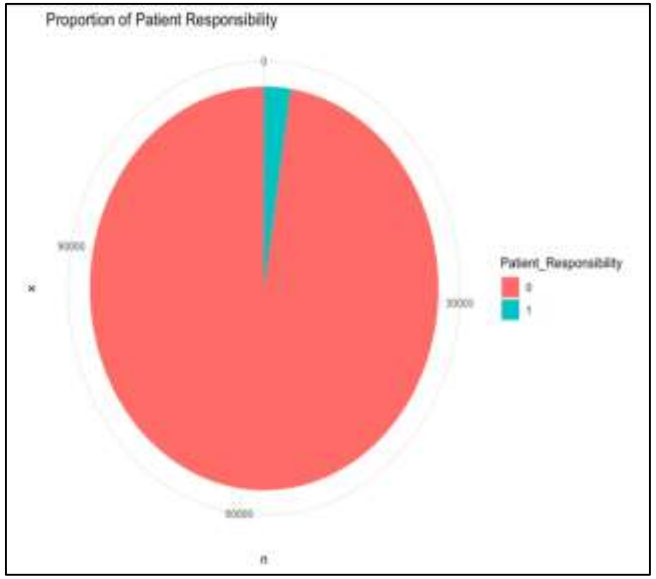
Patients Responsibility:

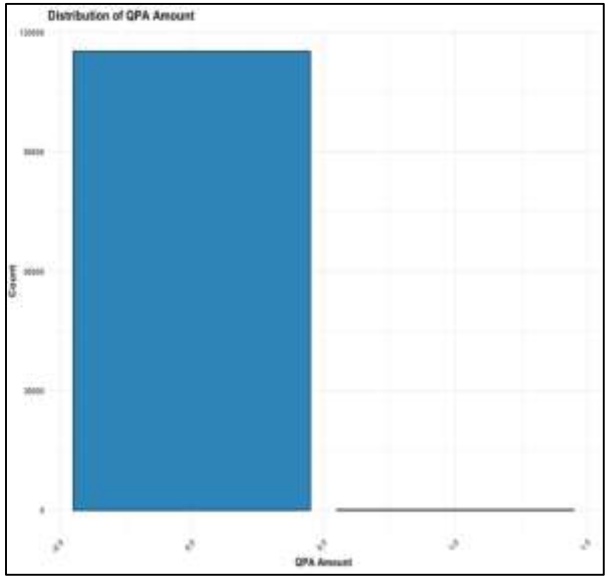
- Over 115,000 instances of Patient Responsibility are at 1.0.
- Very few entries have a value of 0.5.The distribution is highly skewed.
- The majority of Patient Responsibility amounts are uniform.
- Predominantly concentrated at the value of 1.0 with minimal variations.

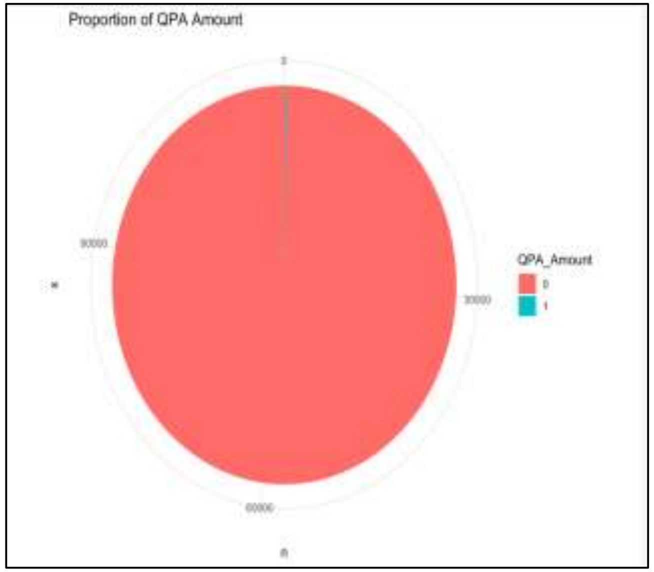
QPA Amount:

- The bar chart shows a highly skewed distribution of QPA values.
- Nearly all entries have a QPA amount of 0.5, with over 115,000 instances, there is a negligible count of entries with a QPA amount of 1.0.
- The majority of QPA amounts are concentrated around the 0.5 value, very few deviations from the 0.5 value.
- QPA values are predominantly uniform across the dataset.



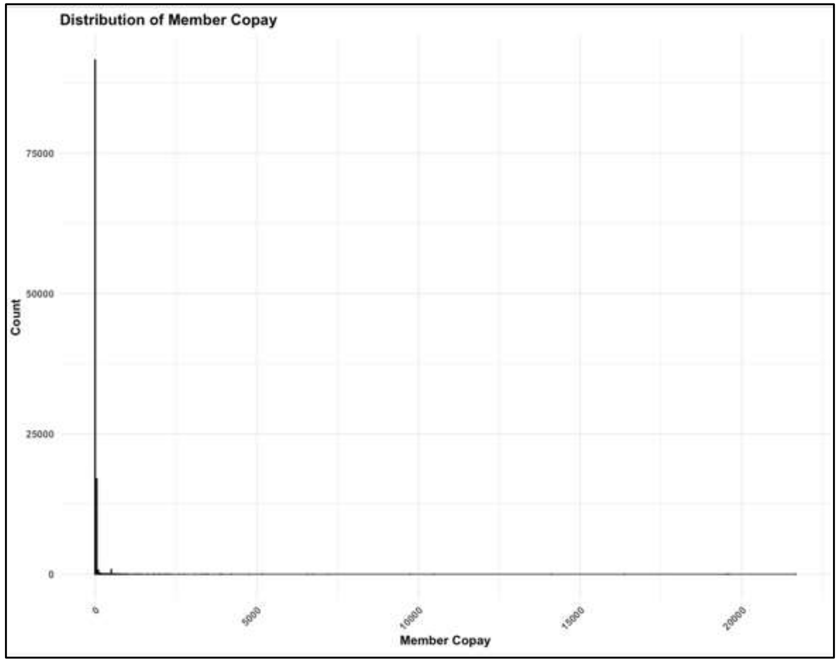


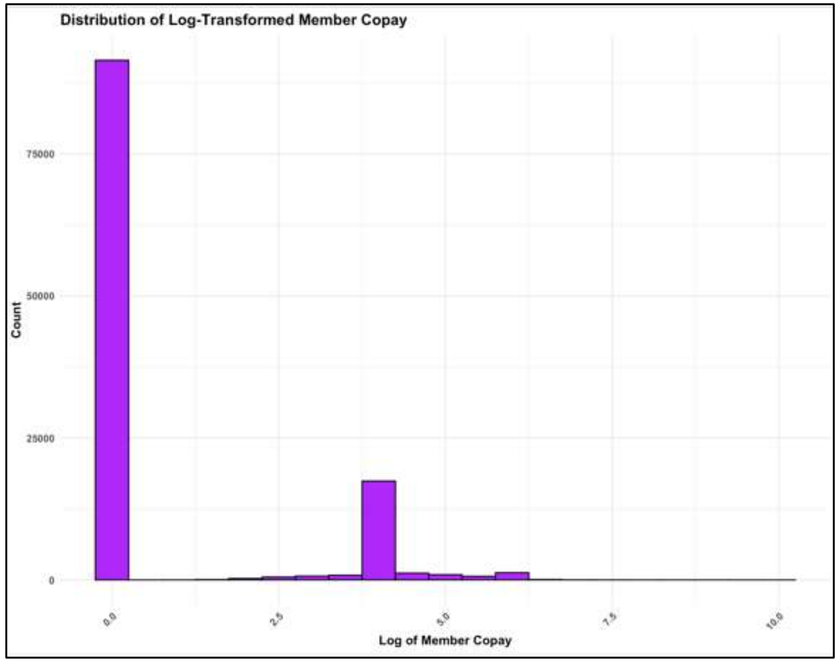




Member Copay:

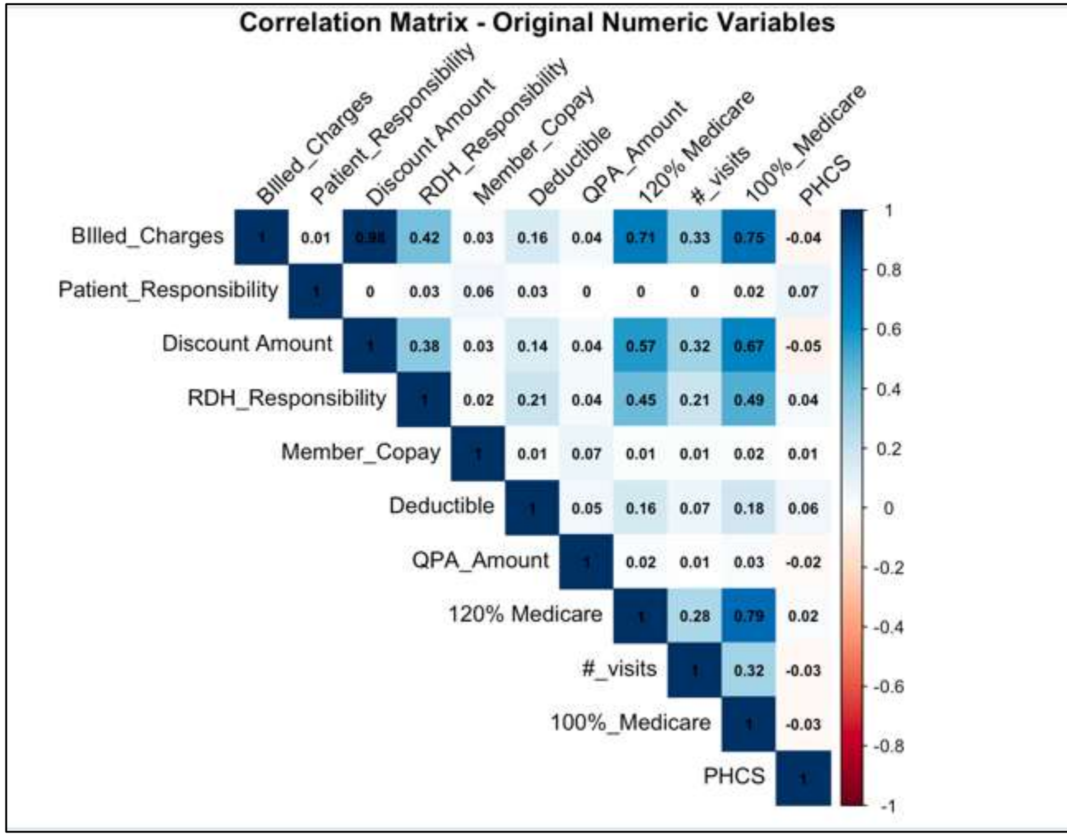
- The initial histogram of Member Copay shows a highly **skewed distribution** with values concentrated around zero.
- A long tail extends towards higher values, indicating significant skewness.
- The skewness makes it challenging to perform statistical analyses that assume normality.
- After log transformation, the distribution becomes more normalized.
- Log transformation reduces skewness, compresses the data range, and highlights differences among smaller values.
- The transformation stabilizes variance, making the data more suitable for statistical modeling and analysis.





Correlation Matrix of Original Numeric Variables

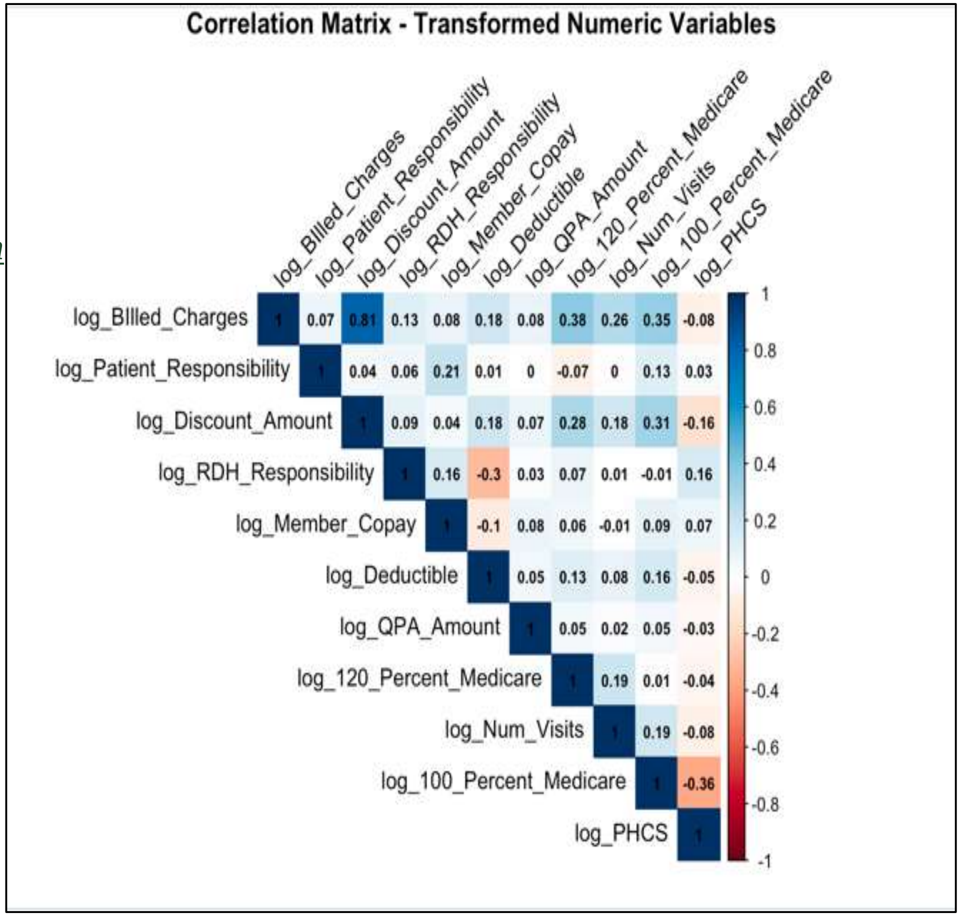
- The correlation matrix shows varying degrees of relationships between variables, with values ranging from -1 to 1.
- Billed Charges are ***highly positively correlated*** with Discount Amount (0.98), 100% Medicare (0.75), and 120% Medicare (0.71).
- Patient Responsibility has ***negligible correlations*** with most variables, indicating independent variation.
- Discount Amount is ***highly positively correlated*** with Billed Charges (0.98), 100% Medicare (0.67), and 120% Medicare (0.57).
- RDH Responsibility shows ***moderate positive correlations*** with Billed Charges (0.42), 100% Medicare (0.49), and 120% Medicare (0.45).
- Member Copay has almost ***zero correlation*** with most variables, reflecting minimal relationships.
- Deductible is ***weakly positively correlated*** with Billed Charges (0.16), RDH Responsibility (0.21), and 100% Medicare (0.18).
- Overall, Billed Charges and Discount Amount show strong correlations with Medicare-related variables, while Patient Responsibility varies independently.



Correlation Matrix of log transformed variables

The correlation matrix for log-transformed numeric variables provides insights into linear relationships after applying log transformation.

- Log_Billed_Charges and Log_Discount_Amount show a very ***strong positive correlation*** (0.98).
- Log_Billed_Charges and Log_100_Percent_Medicare have a ***strong positive correlation*** (0.79).
- Log_120_Percent_Medicare and Log_100_Percent_Medicare also exhibit a ***strong positive correlation*** (0.79).
- Log_Billed_Charges and Log_120_Percent_Medicare have a ***strong positive correlation*** (0.71).
- Log_Discount_Amount is ***moderately correlated*** with Log_100_Percent_Medicare (0.67) and Log_120_Percent_Medicare (0.57).
- Log_Num_Visits shows a ***moderate positive correlation*** with Log_Discount_Amount (0.32) and Log_100_Percent_Medicare (0.32).
- Log_RDH_Responsibility is ***moderately correlate***d with Log_100_Percent_Medicare (0.49) and Log_120_Percent_Medicare (0.45).
- Most other variables, including Log_RDH_Responsibility and Log_PHCS, show ***weak or negligible correlations*** with other variables. Log transformation normalizes the data, revealing clearer relationships, primarily between billed charges, discount amounts, and Medicare-related values.



Principal Component Analysis

Purpose of the Principal Component Analysis

The purpose of the Principal Component Analysis is to simplify the dataset by reducing it to the few principal components that still capture the most variance, including the Billed Charges and the RDH Responsibility.

By using this method, it helps identify the key patterns and improve the efficiency and accuracy of the further analysis and the predictive models.

Exploratory Analysis

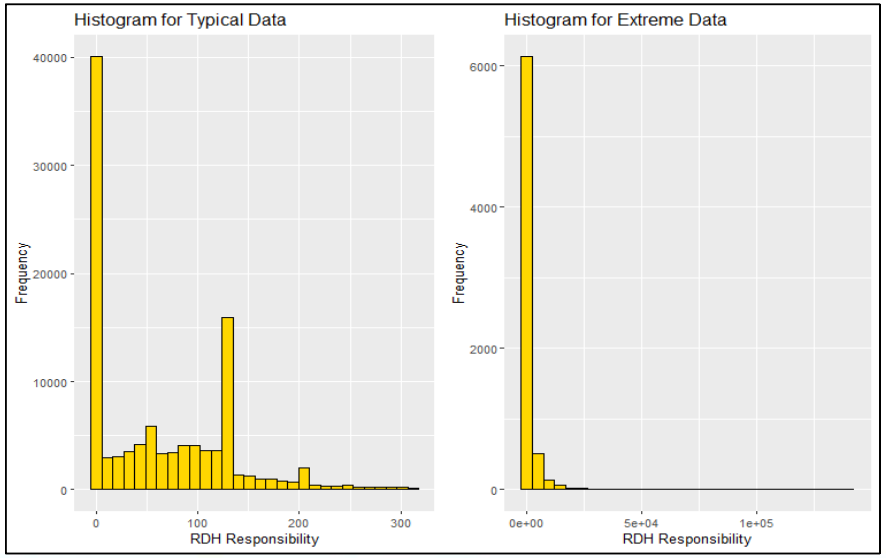
Before removing zero values (Before Data Cleaning)

Typical Data

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	0.00	50.00	62.77	122.57	312.32

Extreme Data

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
312.5	446.5	651.7	1644.2	1147.4	140562.8



Density Plots

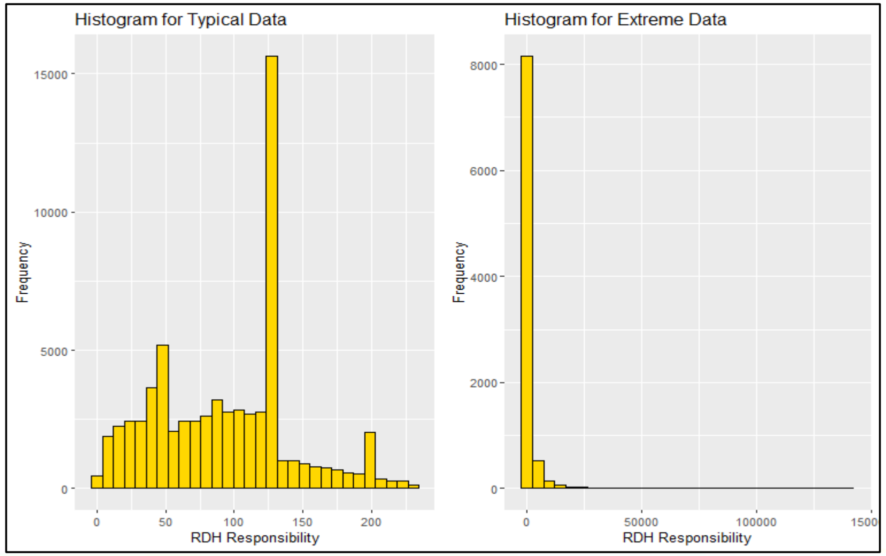
After removing zero values (After Data Cleaning)

Typical Data

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.01	50.00	97.96	93.52	125.00	230.80

Extreme Data

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
230.9	325.4	512.0	1333.6	913.6	140562.8

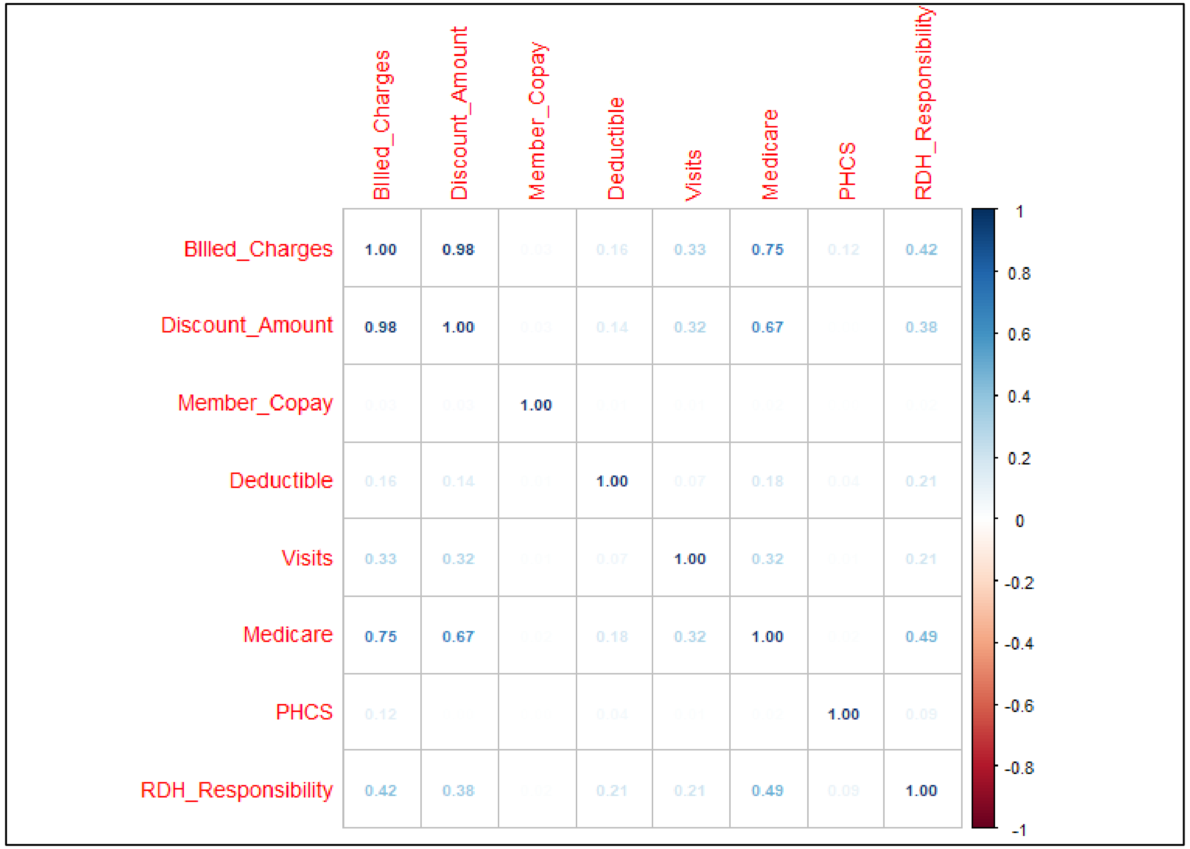


Principal Component Analysis

- There were **115,371** records with **ten** quantitative variables used in this analysis.
- The **Co-Ins (1864)** and **QPA Amount (155)** were removed due to the number of missing values

Quantitative Variables	
BIlled Charges	# visits
Deductible	100% Medicare
Discount Amount	PHCS
Member Copay	RDH Responsibility

- The correlation matrix was created to see the linear relationship between the variables.



Principal Component Analysis

- Used the full data set and applied PCA

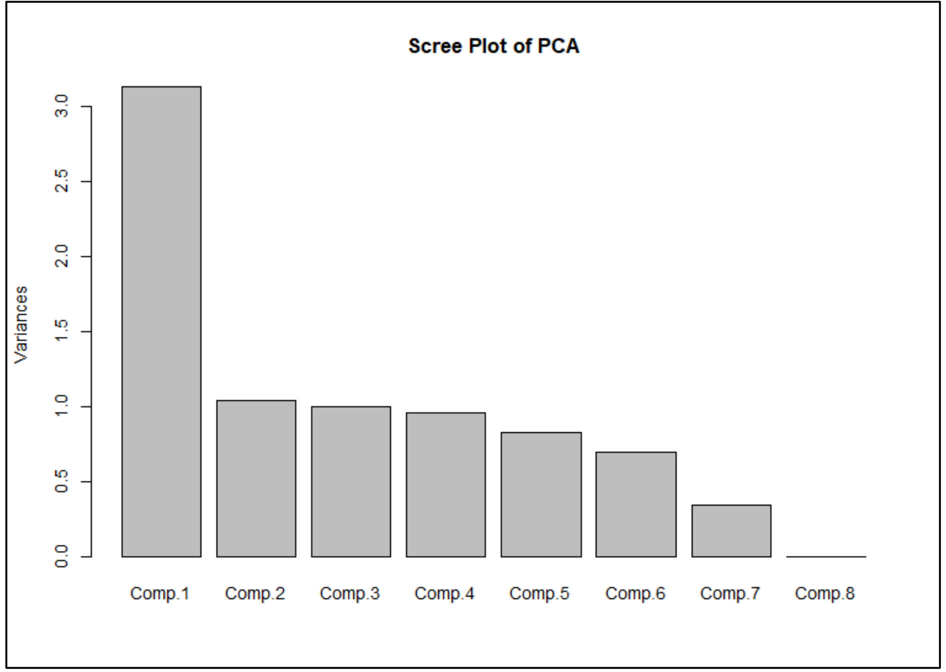
Importance of components:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
Standard deviation	1.7691450	1.0196593	0.9997130	0.9788385	0.9086605	0.83727220	0.5856688	0.0563319098
Proportion of Variance	0.3912343	0.1299631	0.1249283	0.1197656	0.1032080	0.08762809	0.0428760	0.0003966605
Cumulative Proportion	0.3912343	0.5211974	0.6461257	0.7658913	0.8690992	0.95672734	0.9996033	1.0000000000

Loadings:

	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6	Comp.7	Comp.8
BIlled_Charges	0.532			0.152	0.195	0.238	0.199	0.744
Discount_Amount	0.509	0.180			0.214	0.280	0.388	-0.651
Member_Copay			-0.994	0.101				
Deductible	0.158	-0.520		-0.748	-0.104	0.359		
Visits	0.270	0.173			-0.942			
Medicare	0.483				0.113		-0.852	-0.118
PHCS		-0.773		0.605		0.114		
RDH_Responsibility	0.352	-0.244		-0.157		-0.845	0.279	

Scree Plot



Comp.1 = 0.532 (Billed Charges) + 0.509(Discount Amount) + 0.483(Medicare) + 0.352(RDH Responsibility) + 0.270(Visits) + 0.158(Deductible)

Principal Component Analysis

Positively related variables

- Discount_Amount and Billed_Chargers
- Medicare and Billed_Chargers
- Discount_Amount and # visit

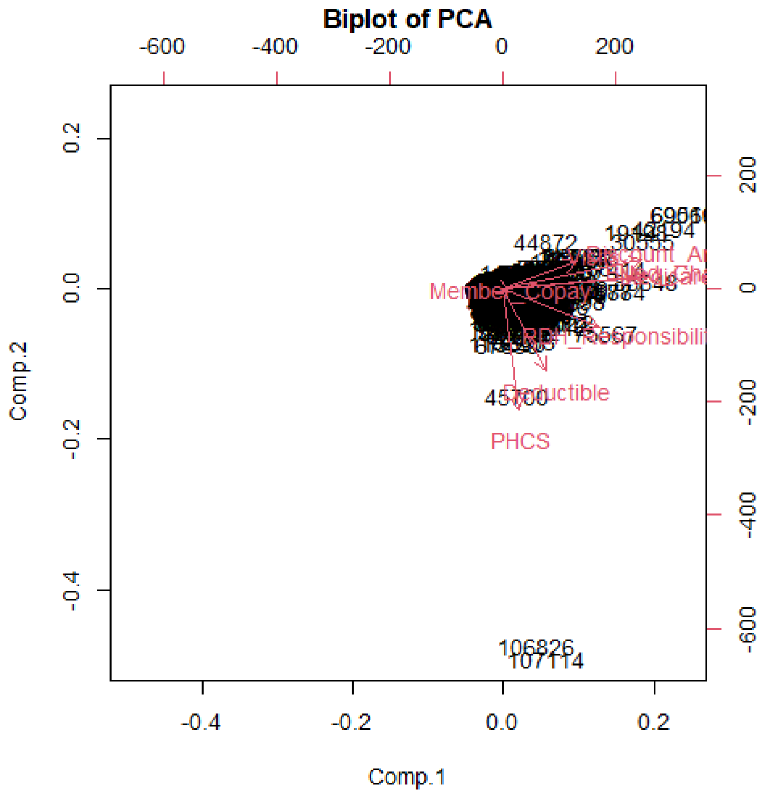
Unrelated variables

- PHCS and Medicare
- PHCS and Billed
- Deductible and # visit

Negatively related variables

- PHCS and # visit

Biplot



Purpose of the Time Plots

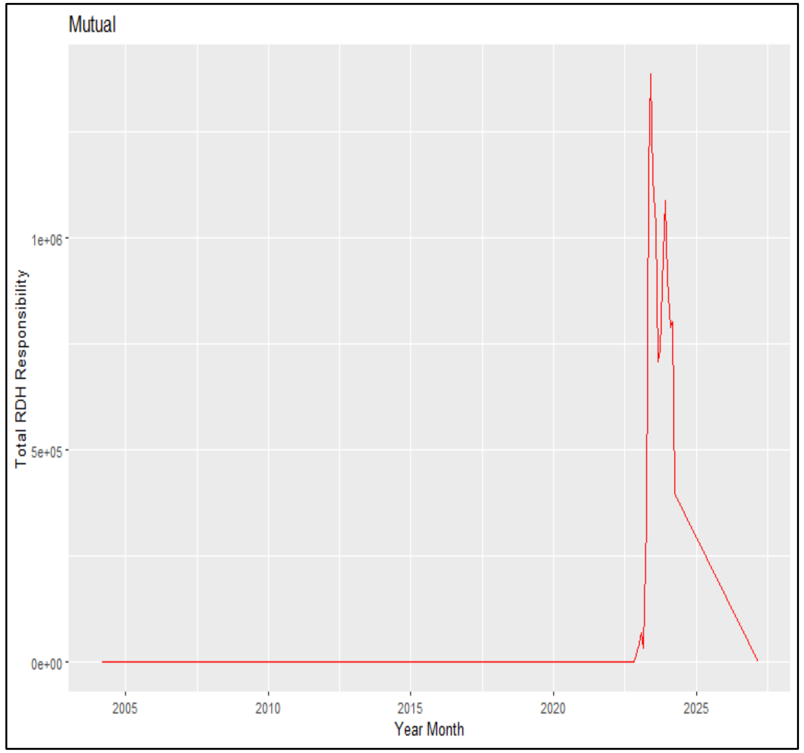
By using time plots, it helps show the RDH Responsibility over time by specific Claim Buckets. It shows the months where higher claims were located, which can help show if this has something to do with a seasonal health issue or insurance policy renewal.

By looking at the data by month to month, it shows the needed area of improvements for RDH Responsibility in the certain Claim Buckets and help locate the possible causation.

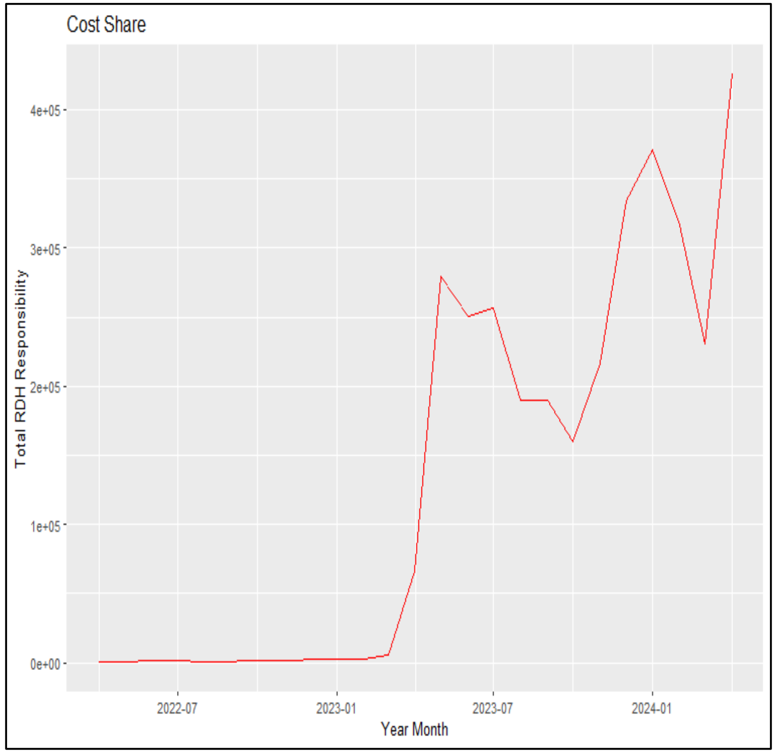
Time Plots

RDH Responsibility vs. Claims Bucket

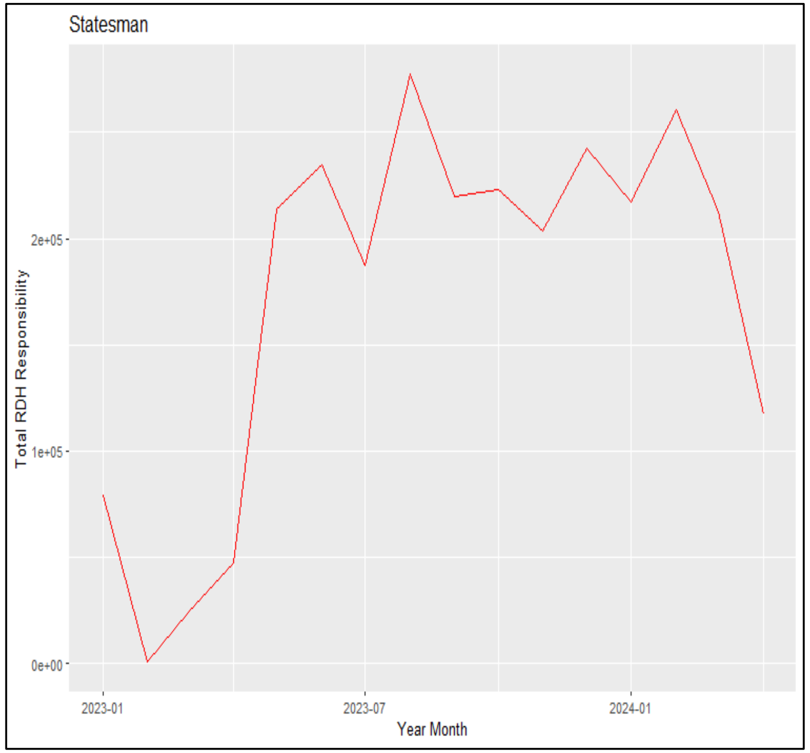
Mutual



Cost Share



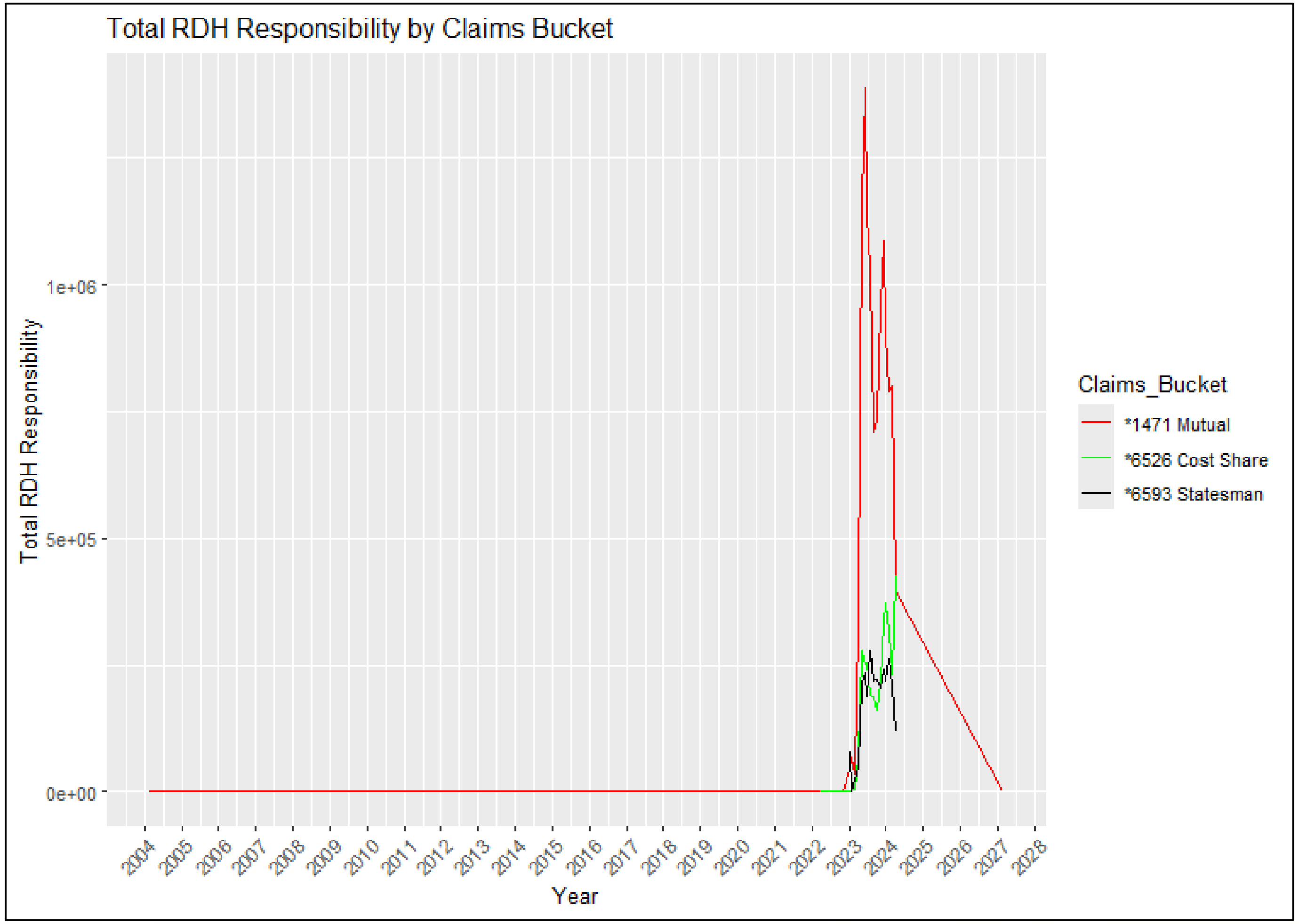
Statesman



Baylor University

Time Plots

RDH Responsibility vs. Claims Bucket



- A significant increase in RDH responsibility begins around late 2022 or early 2023 across all three claims buckets.

- There's a rapid decline in RDH responsibility across all buckets starting in 2024 except Cost Share

Baylor University

Purpose of ANOVA Tests and Tukey HSD Tests

ANOVA Tests

It reveals that the log-transformed ANOVA results are more reliable and interpretable, suggesting that performing ANOVA on log-transformed data is beneficial for this dataset.

Tukey HSD Tests

It reveals significant differences between all the variables and shows the significant impact of billings and savings determined by the other variables within the dataset.

ANOVA Test and Tukey HSD Testing for Coverage Tier

ANOVA Test:

- No significant difference in RDH Responsibility across different Coverage Tiers (F-value: 0.077, p-value: 0.989).

```
> # ANOVA for Coverage_Tier
> anova_coverage_tier <- aov(RDH_Responsibility ~ Coverage_Tier, data = Cleaned_MCL_Data)
> summary(anova_coverage_tier)
                  Df    Sum Sq Mean Sq F value Pr(>F)
Coverage_Tier      4 4.043e+05  101084   0.077  0.989
Residuals     115361 1.505e+11 1304492
```

Tukey HSD Testing:

The results indicate significant differences in log-transformed RDH Responsibility for specific pairs of Coverage Tier categories:

- FAM vs. EMP: FAM has ***higher*** RDH Responsibility.
- ESP vs. FAM: ESP has lower RDH Responsibility.
- ECH vs. EMP: ECH has ***higher*** RDH Responsibility.
- ECH vs. ESP: ECH has ***higher*** RDH Responsibility.

No Significant Differences:

- ESP vs. EMP ECH vs. FAM Others vs. EMP
- Others vs. FAM Others vs. ECH Others vs. ESP

```
> print(tukey_coverage_tier)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ Coverage_Tier, data = Cleaned_MCL_Data)

$Coverage_Tier
                  diff         lwr         upr     p adj
FAM-EMP     0.28394958  0.23870126  0.32919790 0.0000000
ESP-EMP     0.00148169 -0.05341449  0.05637787 0.9999934
ECH-EMP     0.24983688  0.18034572  0.31932804 0.0000000
Others-EMP  0.09901357 -0.24976415  0.44779129 0.9380358
ESP-FAM    -0.28246789 -0.34300599 -0.22192979 0.0000000
ECH-FAM    -0.03411270 -0.10814170  0.03991630 0.7176478
Others-FAM -0.18493601 -0.53464613  0.16477411 0.6000306
ECH-ESP     0.24835519  0.16806526  0.32864512 0.0000000
Others-ESP  0.09753188 -0.25355693  0.44862068 0.9425340
Others-ECH -0.15082331 -0.50448810  0.20284147 0.7723518
```

Baylor University

ANOVA Test and Tukey HSD Testing for Claims Bucket

ANOVA Test:

- Significant difference in RDH Responsibility across different Claims Buckets (F-value: 107.6, p-value: < 2e-16 ***).

```
> # ANOVA for Claims_Bucket
> anova_claims_bucket <- aov(RDH_Responsibility ~ Claims_Bucket, data = Cleaned_MCL_Data)
> summary(anova_claims_bucket)
                  Df    Sum Sq   Mean Sq F value Pr(>F)
Claims_Bucket      3 4.201e+08 140017674   107.6 <2e-16 ***
Residuals     115362 1.501e+11   1300843
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey HSD Testing:

The results indicate significant differences in log-transformed RDH Responsibility for various pairs of Claims Bucket categories.

- *Mutual vs. Cost Share:* Mutual has a slightly lower RDH Responsibility.
- *Others vs. Cost Share:* Others have a much lower RDH Responsibility.
- *Statesman vs. Cost Share:* Statesman has a ***higher*** RDH Responsibility.
- *Others vs. Mutual:* Others have a significantly lower RDH Responsibility.
- *Statesman vs. Mutual:* Statesman has a ***higher*** RDH Responsibility.
- *Statesman vs. Others:* Statesman has a significantly ***higher*** RDH Responsibility.

```
> print(tukey_claims_bucket)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ Claims_Bucket, data = Cleaned_MCL_Data)

$Claims_Bucket
                          diff        lwr         upr     p adj
Mutual-Cost Share  -0.06479874 -0.1081066 -0.02149091 0.0006993
Others-Cost Share  -3.24288696 -3.2996800 -3.18609390 0.0000000
Statesman-Cost Share 0.18306864  0.1235377  0.24259954 0.0000000
Others-Mutual      -3.17808822 -3.2247509 -3.13142557 0.0000000
Statesman-Mutual    0.24786737  0.1979086  0.29782617 0.0000000
Statesman-Others    3.42595560  3.3639414  3.48796982 0.0000000
```

Baylor University

ANOVA Test and Tukey HSD Testing for Status

ANOVA Test:

- Significant difference in RDH Responsibility across different Status categories (F-value: 82.56, p-value: < 2e-16 ***).

```
> # ANOVA for Status
> anova_status <- aov(RDH_Responsibility ~ Status, data = Cleaned_MCL_Data)
> summary(anova_status)
                Df    Sum Sq   Mean Sq F value Pr(>F)
Status           3 3.224e+08 107467793   82.56 <2e-16 ***
Residuals   115362 1.502e+11   1301689
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey HSD Testing:

Status is a critical factor affecting RDH Responsibility.

- *Complete:* Generally, has ***higher*** RDH Responsibility compared to other statuses.
- *Duplicate:* Has significantly lower RDH Responsibility compared to Complete and Open.
- *Open:* Shows ***higher*** responsibility when compared to Duplicate but lower when compared to Complete.
- *Deny:* Has the lowest RDH Responsibility when compared to Complete and Open.

```
> print(tukey_status)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ Status, data = Cleaned_MCL_Data)

$Status
                         diff        lwr        upr     p adj
Duplicate-Complete -3.1389641 -3.1959149 -3.0820133 0.0000000
Open-Complete      -0.7030366 -0.7946957 -0.6113776 0.0000000
Deny-Complete      -3.3335005 -3.8559963 -2.8110048 0.0000000
Open-Duplicate      2.4359274  2.3308533  2.5410015 0.0000000
Deny-Duplicate     -0.1945365 -0.7195517  0.3304787 0.7766888
Deny-Open          -2.6304639 -3.1603686 -2.1005592 0.0000000
```

ANOVA Test and Tukey HSD Testing for Provider Type

ANOVA Test:

- Significant difference in RDH Responsibility across different Provider Types (F-value: 255.5, p-value: < 2e-16 ***).

```
> # ANOVA for Provider_Type
> anova_provider_type <- aov(RDH_Responsibility ~ Provider_Type, data = Cleaned_MCL_Data)
> summary(anova_provider_type)
                  Df    Sum Sq   Mean Sq F value Pr(>F)
Provider_Type      4 1.322e+09 330386587   255.5 <2e-16 ***
Residuals     115361 1.492e+11   1293039
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey HSD Test:

Provider Type is a critical factor affecting RDH Responsibility.

- *Specialist vs. Others:* Specialists have lower RDH Responsibility.
- *Hospital vs. Others:* Hospitals have much lower RDH Responsibility.
- *Labs vs. Others:* Labs have lower RDH Responsibility.
- *ER vs. Others:* ER has lower RDH Responsibility.
- *Hospital vs. Specialist:* Hospitals have much lower RDH Responsibility.
- *Labs vs. Specialist:* Labs have lower RDH Responsibility.
- *ER vs. Specialist:* ER has lower RDH Responsibility.
- *Labs vs. Hospital:* Labs have ***higher*** RDH Responsibility.
- *ER vs. Hospital:* ER has ***higher*** RDH Responsibility.
- *ER vs. Labs:* ER has lower RDH Responsibility.

```
> print(tukey_provider_type)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ Provider_Type, data = Cleaned_MCL_Data)

$Provider_Type
                          diff        lwr        upr p adj
Specialist-Others   -0.5292932 -0.5763424 -0.4822440     0
Hospital-Others     -2.4434155 -2.4938035 -2.3930276     0
Labs-Others         -0.9585270 -1.0109498 -0.9061042     0
ER-Others           -1.6097674 -1.6822135 -1.5373212     0
Hospital-Specialist -1.9141224 -1.9651568 -1.8630879     0
Labs-Specialist     -0.4292339 -0.4822783 -0.3761894     0
ER-Specialist       -1.0804742 -1.1533714 -1.0075769     0
Labs-Hospital        1.4848885  1.4288614  1.5409156     0
ER-Hospital          0.8336482  0.7585527  0.9087436     0
ER-Labs             -0.6512403 -0.7277160 -0.5747647     0
```

ANOVA Test and Tukey HSD Testing for Pricing Strategy

ANOVA Test:

- Significant difference in RDH Responsibility across different Pricing Strategies (F-value: 785, p-value: < 2e-16 ***).

```
> # ANOVA for Pricing Strategy
> anova_pricing_strategy <- aov(RDH_Responsibility ~ Pricing_Strategy, data = Cleaned_MCL_Data)
> summary(anova_pricing_strategy)
                     Df    Sum Sq   Mean Sq F value Pr(>F)    
Pricing_Strategy      3 3.010e+09 1.003e+09     785 <2e-16 ***
Residuals        115362 1.475e+11 1.278e+06                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey HSD Testing:

Pricing Strategy is a critical factor affecting RDH Responsibility.

- *Medicare:* Generally, has ***higher*** RDH Responsibility compared to other pricing strategies.
- *PHCS:* Shows significantly lower RDH Responsibility compared to Medicare, Others, and QuickPay.
- *QuickPay:* Has lower responsibility compared to Medicare and Others, but higher compared to PHCS.
- *Others:* Shows a moderate level of responsibility, ***higher*** than PHCS but lower than Medicare.

```
> print(tukey_pricing_strategy)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ Pricing_Strategy, data = Cleaned_MCL_Data)

$Pricing_Strategy
                        diff        lwr        upr    p adj
Others-Medicare   -0.1065806 -0.1849838 -0.0281773 0.002697
PHCS-Medicare     -1.1858485 -1.2444017 -1.1272953 0.000000
Quickpay-Medicare -0.8737404 -0.9361872 -0.8112936 0.000000
PHCS-Others       -1.0792680 -1.1415775 -1.0169585 0.000000
Quickpay-Others   -0.7671598 -0.8331416 -0.7011780 0.000000
Quickpay-PHCS      0.3121081  0.2716729  0.3525434 0.000000
```

ANOVA Test and Tukey HSD Testing for Pre-Authorization

ANOVA Test:

- Significant difference in RDH Responsibility across different Pre-Authorization statuses (F-value: 486.7, p-value: < 2e-16 ***).

```
> # ANOVA for Pre Authorization
> anova_pre_authorization <- aov(RDH_Responsibility ~ `Pre-Authorization`, data = Cleaned_MCL_Data)
> summary(anova_pre_authorization)
                       Df    Sum Sq   Mean Sq F value Pr(>F)    
`Pre-Authorization`     2 1.259e+09 629542915   486.7 <2e-16 ***
Residuals          115363 1.492e+11   1293558                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey HSD Testing:

The results indicate significant differences in log-transformed RDH Responsibility for certain pairs of Pre-Authorization categories.

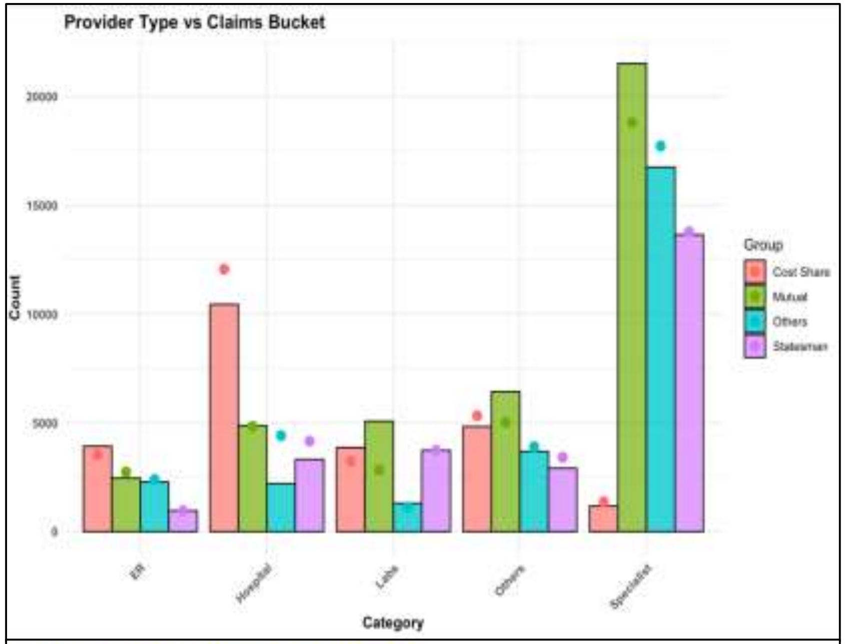
- *No vs. Yes:* Significant difference, with the Yes category showing ***higher*** log-transformed RDH Responsibility.
- *Unknown vs. Yes:* Significant difference, with the Yes category showing ***higher*** log-transformed RDH Responsibility.
- *Unknown vs. No:* No significant difference in log-transformed RDH Responsibility.

```
> print(tukey_pre_authorization)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = log_RDH_Responsibility ~ PreAuthorization, data = Cleaned_MCL_Data)

$PreAuthorization
                   diff        lwr        upr     p adj
Unknown-No  -0.07732222 -0.1712996 0.01665516 0.1307118
Yes-No       1.59061294  1.4851812 1.69604464 0.0000000
Yes-Unknown  1.66793516  1.6145424 1.72132787 0.0000000
```

Chi-Square Test

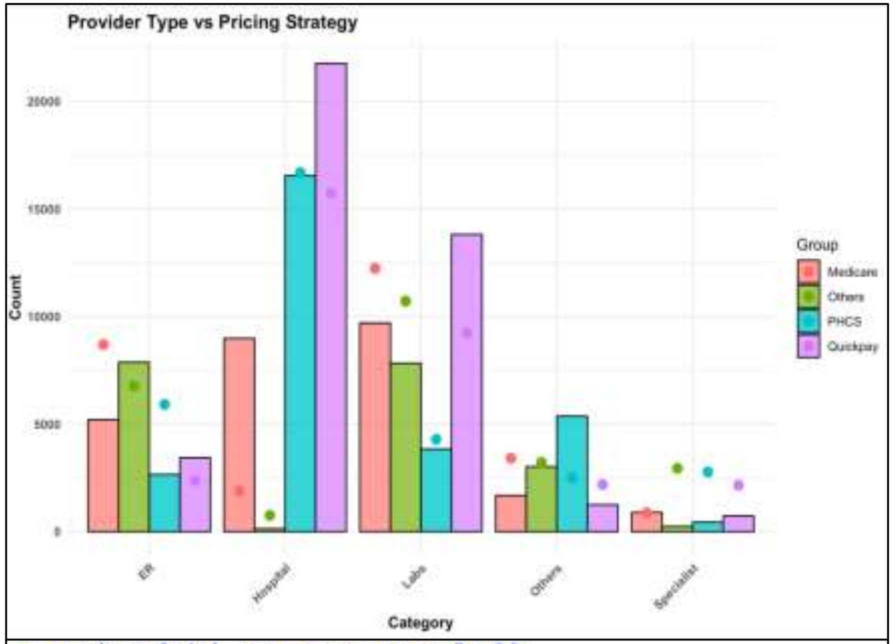



```
> print(chi_square_result1)

	Pearson's Chi-squared test

data:  provider_vs_bucket
X-squared = 4504.4, df = 12, p-value < 2.2e-16
```

It shows that the extremely low p-values suggest that there is statistically significant association between those two variables. The high chi-square value means the distributions do not have a good association between them. This is also an important correlation technique to see which variables are important within the data.




```
> print(chi_square_result3)

	Pearson's Chi-squared test

data:  provider_vs_pricing
X-squared = 46917, df = 12, p-value < 2.2e-16
```

Baylor University

Purpose of Cluster Analysis and Centers

Cluster Analysis

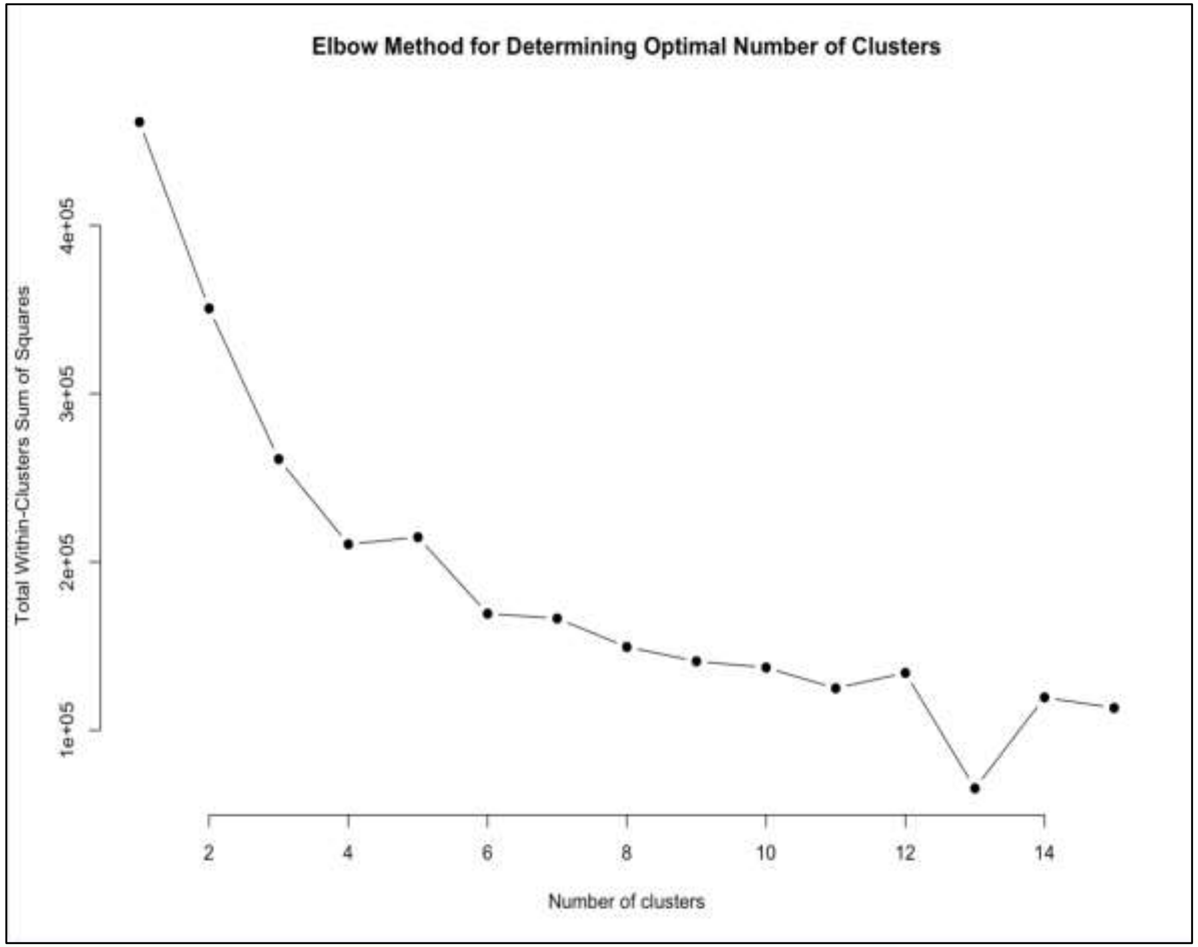
It helps build the distinct groups based on the similarities between the Discount Amount and Billed Charges and highlights the similar patterns to inform where members fall in the prices of Billed Charges and Discount Amount.

Cluster Centers

The cluster centers help build an average of the clusters to show where majority of the similar charges and amount patterns fall to show the differences between the groups amount in their Billed Charges and Discount Amount.

Cluster Analysis

- WSS ***decreases sharply*** as the number of clusters increases from 2 to around 6, indicating effective grouping.
- After around 6 clusters, the rate of decrease in WSS slows significantly, suggesting diminishing returns from adding more clusters.
- The ***elbow point*** appears around ***6 clusters***, where the curve bends, and the rate of WSS decrease slows.
- Use of 6 clusters for k-means clustering to balance model complexity and clustering quality



Baylor University

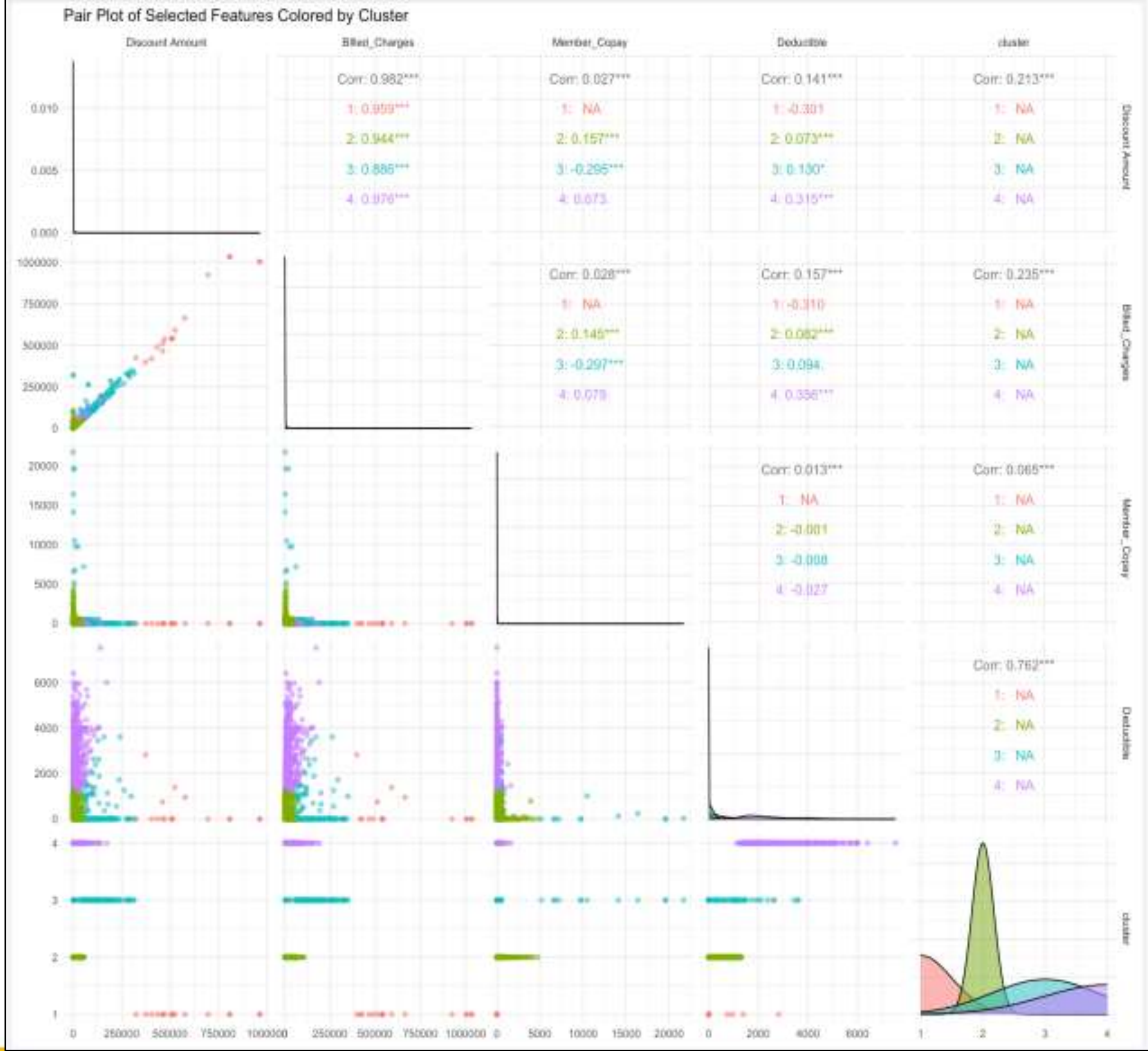
Cluster Analysis; Pair Plot

- Clusters are clearly distinguished by different colors, indicating ***successful separation*** of data points into distinct groups.

- *Discount Amount vs. Billed Charges:* ***Strong positive correlation*** between Discount Amount and Billed Charges across all clusters (e.g., 0.982 overall, 0.959 for Cluster 1). Higher billed charges tend to have higher discount amounts.
- *Member Copay:* ***Very low correlation*** with other features, indicating relative independence from Billed Charges, Discount Amount, and Deductible.
- *Deductible:* ***Positive correlation*** with Discount Amount and Billed Charges in some clusters.

- *Cluster 1 (Red):* ***High correlation*** between Discount Amount and Billed Charges.
- *Cluster 2 (Green):* ***Moderate to high correlation*** between Discount Amount and Billed Charges.
- *Cluster 3 (Blue):* ***High correlation*** between Discount Amount and Billed Charges, with some correlation with Deductible.
- *Cluster 4 (Purple):* ***High correlation*** between Discount Amount and Billed Charges, similar to other clusters.

- Density plots on the diagonal show feature distribution within each cluster. Deductible shows more spread-out distribution in Cluster 4, indicating variability within this cluster.



Baylor University

Cluster Summary

Cluster 1:

- This cluster has the ***highest mean*** values for both Discount Amount and Billed Charges, indicating instances with significantly higher charges and discounts. Member Copay is 0, suggesting no copay for members in this cluster, while the deductible is moderately high. Represents ***high-cost claims*** with no copay and moderate deductible.

Cluster 2:

- This cluster has the ***lowest mean*** values for Discount Amount and Billed Charges, indicating low-cost claims. The Member Copay and Deductible are also very low, suggesting minimal financial responsibility for members in this cluster. Represents ***low-cost claims*** with minimal member financial responsibility.

Cluster 3:

- This cluster represents ***moderate*** Discount Amount and Billed Charges values. It also has a higher Member Copay compared to Clusters 1 and 2, suggesting members have a more significant financial responsibility. The deductible is moderately low. Represents ***moderate-cost claims*** with higher copay and moderate deductible.

Cluster 4:

- This cluster has ***low to moderate*** Discount Amount and Billed Charges values. The Member Copay is relatively low, but the deductible is extremely high compared to other clusters, indicating that members in this cluster must pay a large amount out-of-pocket before insurance coverage is active. Represents ***low to moderate-cost claims*** with low copay but very high deductible.

```
> print(cluster_summary)
# A tibble: 4 × 5
  cluster `Discount Amount` BIlled_Charges Member_Copay Deductible
    <int>             <dbl>          <dbl>        <dbl>      <dbl>
1       1           576313.        654135.          0        347. 
2       2              688.           974.         17.7       23.5
3       3            99456.        119563.        502.       182. 
4       4            15276.         20941.         41.7     2521. 
```

Baylor University

Cluster Centers

Cluster 1:

- Highest values for Discount Amount and Billed Charges.
- Very low or zero Member Copay.
- Low deductible.

Cluster 2:

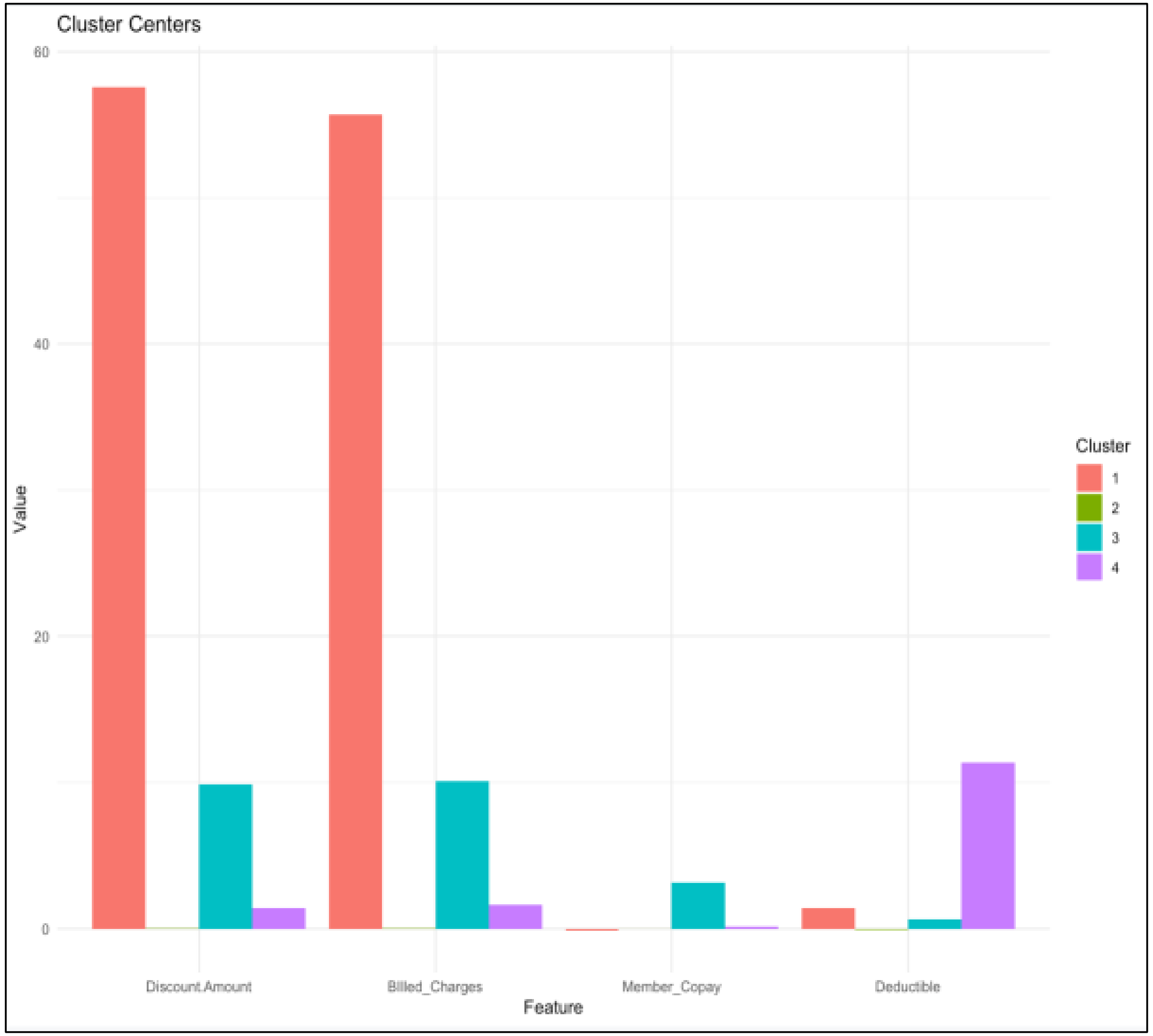
- Lowest values for all features, indicating minimal financial amounts involved.

Cluster 3:

- Moderate values for Discount Amount and Billed Charges.
- Moderate Member Copay.
- Low deductible.

Cluster 4:

- Low values for Discount Amount and Billed Charges.
- Low Member Copay.
- Highest deductible.



Baylor University

Final Model

Logistic Regression

Purpose of Logistic Regression

This regression highlights the valuable insights on the binary outcomes between the variables within the dataset. By using this, it also provides predictive insights to help drive data-driven decision making. With data-driven decision making, it can help optimize resource allocation within Redirect Health.

Exploratory Analysis

- There were **115,366** records with **five** qualitative and **twelve** quantitative predictor variables used in this analysis.
- The target variable **("Status")** was transformed into a binary variable, where 1 indicates "complete" and 0 indicates otherwise.
- The **Likelihood Ratio Test** was conducted to assess the overall significance of each predictor variable.

Qualitative Predictors	**Quantitative Predictors**
Provider Type...6	Billed Charges
Claims Bucket	Discount Amount (Saving amount)
Pricing Strategy	Member Copay
Coverage Tier	Deductible
Pre-Authorization	# Visits
	100% Medicare
	Patient_Responsibility
	QPA_Amount
	120% Medicare
	PHCS
	RDH_Responsibility
	QPA_Amount

Baylor University

Exploratory Analysis

- The “**Likelihood Ratio Test**" test was applied to each categorical variable to assess statistical significance.
- All categorical variable were statistically significance
- The **Wald Test** was conducted to each quantitative variable to assess statistical significance.
- The "**Member_Copay**" and "**QPA_Amount**" were not statistically significance and were removed.
- Final data set contained 115,366 records with 15 variables.

Logistic Regression Model

Confusion Matrix Based on 0.5 Cut of Value

```
          Reference
Prediction     0     1
         0  7463  3118
         1  6804 97981
```

- **Sensitivity** is 0.97
- **Specificity** is 0.52

```
Call:
glm(formula = formula_str, family = binomial(link = logit), data = MCL_data_lg)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)    
(Intercept)                 2.040e+00  9.310e-01   2.191 0.028456 *  
Provider_TypeHospital      -1.085e-01  5.220e-02  -2.079 0.037623 *  
Provider_TypeLabs          -1.991e+00  5.308e-02 -37.504  < 2e-16 ***
Provider_TypeOthers        -2.356e-01  5.144e-02  -4.581 4.62e-06 ***
Provider_TypeSpecialist     2.486e-01  5.115e-02   4.861 1.17e-06 ***
Claims_BucketMutual         1.310e-01  3.979e-02   3.291 0.000998 ***
Claims_BucketOthers        -2.597e+00  4.050e-02 -64.122  < 2e-16 ***
Claims_BucketStatesman     -2.069e-01  5.173e-02  -3.999 6.37e-05 ***
Pricing_StrategyOthers      1.689e+00  7.738e-02  21.830  < 2e-16 ***
Pricing_StrategyPHCS       -8.269e-01  5.759e-02 -14.359  < 2e-16 ***
Pricing_StrategyQuickpay    1.054e+00  6.989e-02  15.084  < 2e-16 ***
Pre_AuthorizationUnknown   -1.144e+00  1.055e-01 -10.845  < 2e-16 ***
Pre_AuthorizationYes        2.017e-01  1.257e-01   1.604 0.108679    
Coverage_TierEMP           -2.336e-01  4.535e-02  -5.152 2.58e-07 ***
Coverage_TierESP           -1.274e-01  5.198e-02  -2.450 0.014274 *  
Coverage_TierFAM           -2.083e-01  4.795e-02  -4.345 1.40e-05 ***
Coverage_TierOthers        -4.223e-01  2.415e-01  -1.749 0.080290 .  
BIlled_Charges             -6.343e-05  1.094e-05  -5.796 6.80e-09 ***
Discount_Amount             5.746e-05  1.110e-05   5.179 2.24e-07 ***
Medicare_120                3.548e-04  2.295e-05  15.458  < 2e-16 ***
Medicare_100               -3.650e-04  2.324e-05 -15.706  < 2e-16 ***
Deductible                 -1.938e-04  5.887e-05  -3.292 0.000996 ***
Visits                     -1.519e-02  7.192e-03  -2.111 0.034746 *  
Patient_Responsibility      2.794e+00  9.247e-01   3.021 0.002519 ** 
PHCS                       -1.550e-05  2.229e-06  -6.955 3.53e-12 ***
RDH_Responsibility          6.557e-05  1.466e-05   4.473 7.71e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 86333  on 115365  degrees of freedom
Residual deviance: 51077  on 115340  degrees of freedom
AIC: 51129

Number of Fisher Scoring iterations: 7
```

Logistic Regression Model (2nd Method)

- The Data split into ***75%-25%*** training and test sets.
- Utilized ***5-fold cross-validation*** and applied a ***Lasso penalty*** to prevent overfitting.
- The training data accuracy was ***0.913*** and the testing data accuracy was ***0.914***.
- Results indicated that the model was ***well-generalized***, showing neither underfitting nor overfitting.

Confusion Matrix (Train)

Prediction	Truth 0	Truth 1
0	5644	2498
1	5056	73326

Confusion Matrix (Test)

Prediction	Truth 0	Truth 1
0	1921	836
1	1646	24439

Purpose of the Final Model

Purpose of Final Model

The ***high sensitivity*** of the model, 0.9692, shows its ***strong performance*** in identifying ***positive cases***. The purpose of splitting the model into training and testing sets is to provide a more ***realistic measure*** of the performance of the model and helps prevent the overfitting of the model, which helps make the results obtained from the ***model more reliable***.

Through running multiple different models of the data, we have gotten similar results on the model. With high AIC values showing in each model, we have tested; it highlights the ***model's complexity*** and that it is hard to capture the ***underlying patterns*** within the data.

Thank you